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Name	Type of File	Size	Page
MigrationBench_R2R.pdf	Author Response	357.9 KB	Page 3
MigrationBench_main.pdf	Main Document - LaTeX PDF	8.6 MB	Page 15
MigrationBench_SI_LaTeX_source.zip	LaTeX Supplementary File	567.7 KB	*Not converted to PDF*
MigrationBench_SI.pdf	Supporting Information for Peer Review and Publication	821.6 KB	Page 39
MigrationBench-gragh-abstract.pdf	Graphical Abstract Image	2.8 MB	Page 55
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ADVANCED INTELLIGENT SYSTEMS

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Research Article

When Does a Foundation Model Help? Process-Specific Reliability of Machine-Learned Interatomic Potentials

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Response to Reviewer

When Does a Foundation Model Help?
Process-Specific Reliability of Machine-Learned Interatomic Potentials

We thank the reviewers for their careful evaluation, which has strengthened the physical and methodological depth of this manuscript. The paper is now reorganized around seven main figures to establish a framework for evaluating process-specific reliability in foundation models. The revised comparative analysis separates pointwise errors, fixed-image energy assignments, model-relaxed pathways, and macroscopic MD horizons. This decoupling demonstrates the specific regimes where target adaptation provides advantages versus where zero-shot foundation priors remain robust.

To ensure consistency and reproducibility, we have standardized the quantitative metrics using complete grouped retraining. Consequently, the legacy 0.16 eV and 0.41 eV estimates are replaced by these controlled benchmarks. The 89-frame control is now defined as a data-volume sensitivity baseline. Furthermore, the updated MD trajectories capture the failure horizons across all fine-tuned runs, establishing a realistic bound on dynamic stability. We have also refined the feature representation analyses to serve as macroscopic structural diagnostics, mapping the geometric phase space without overextending to causal mechanistic claims. Finally, absolute kinetic interpretations are now constrained to fully converged DFT references, defining the macroscopic boundaries of the workflow.

The point-by-point responses below address each reviewer comment, detailing the specific methodological updates and their locations in the revised text. Within the marked manuscript, the revisions corresponding to these responses are highlighted in blue and annotated with [R1-A1] through [R1-C13] identifiers. All section, figure, and table numbers in this document refer to the revised main text and Supplementary Information.

A1. Fig. 3d versus Fig. S3: NEB barrier discrepancy

> *"Reconcile the foundation-model in-gap barrier discrepancy: state explicitly whether Fig. 3d (fixed-geometry, ∼0.7 eV overestimate, 'unacceptable') and Fig. S3 (self-run NEB, 0.41 eV, 'exceptional') use the same pathway and DFT reference; fix the reference value (0.34 vs. ∼0.3 eV); present both evaluations side-by-side and explain why relaxation changes the verdict. As written, the paper's central ranking is ambiguous."*

Response. We sincerely thank the reviewer for pointing this out. We agree that the original presentation was ambiguous, and we have restructured the migration pathway evaluation to ensure geometric and computational consistency.

The discrepancy in the original manuscript arose from comparing different physical tasks evaluated on distinct structural sets. Specifically, the 0.34 eV reference originates from a 61-atom, single-Cr, 5-image geometry (legacy G1 set). The 0.41 eV record was derived from a larger 82-atom, dual-Cr configuration evaluated with different electronic structure calculators. To establish a standardized benchmark, we have removed these legacy records and now use the 81-atom canonical in-gap reference (1.52 eV).

A data audit has further standardized the evaluation matrix. We explicitly classify the precise topological states and their corresponding energy peaks (e.g., the structural optimization of Scratch-600K seed 123/within-QL self-establishes a finite peak of 2.35 eV). These structural evaluations are strictly verified via SHA256 model identities and are explicitly detailed in the revised main figures.

To clarify the methodology, the revised manuscript separates fixed-geometry calibration from model-driven path generation, presenting both profiles side-by-side in main Fig. 4a–f. Because self-relaxed NEB alters the sampled coordinates, predicting a lower energy peak indicates the model

has navigated to a distinct local basin or pathway. Validating a physical barrier requires structural correspondence with the canonical first-principles trajectory.

Table 1: **Migration-path reference sets standardized and separated.** The values represent forward peaks on distinct image sets, demonstrating that peak energies across different structures cannot serve as interchangeable barrier estimates.

Reference set	Atoms	Images	Forward peak (eV)	Validation status
Legacy G1	61	5	0.34	Baseline diagnostic; restricted to legacy control
Canonical in-gap	81	10	1.52	Diagnostic reference; primary canonical benchmark
Discarded record	82	12	0.41	Calculator mismatch; excluded from benchmark

Takeaway. The evaluation framework has been updated to remove legacy data. The revised analysis separates static energy calibration from dynamic path generation. Macroscopic kinetic claims are now constrained to structurally verified, canonical reference pathways.

Where to find the change. Main Secs. 2.3 and 5.2; Fig. 4a–f and Fig. 5b–e.

Table 1; SI Secs. 2.2.

A2. Train/test overlap for FT-600K

> *"Rule out train/test overlap for FT-600K: report the minimum SOAP-distance (or RMSD) between each in-gap NEB image and its nearest fine-tuning frame, or re-run FT-600K with the NEB-adjacent trajectory segment excluded, and confirm the 0.16 eV result survives."*

Response. To address this, we have updated the data partitioning protocol. The revised grouping mechanism now assigns complete, contiguous source trajectories to either the train, validation, or test sets. This prevents adjacent-frame leakage between partitions and establishes a 600 K dataset split of 1,786/311/267 configurations.

A structural audit confirms the isolation of the 25 legacy NEB images from the training distribution. The audited training files contain zero exact global or Cr-local matches. Minimum SOAP cosine distances remain above the 10^{-4} threshold (5.4×10^{-4} for the full training set and 6.4×10^{-4} for the 89-frame subset). Additionally, the minimum composition-compatible local RMSD is 2.06 Å, exceeding the 0.05 Å near-duplicate threshold. This explicit SOAP and RMSD check confirms the structural independence of the audited image set.

Under this trajectory-split evaluation, the grouped FT-600K G1 discrepancy is 0.47 ± 0.12 eV across three independent training seeds. This multi-seed benchmark replaces the preliminary 0.16 eV single-run estimate. Furthermore, these legacy G1 controls are kept separate from the canonical pathways to maintain independent validation sets for subsequent macroscopic evaluations.

Where to find the change. Main Sec. 5.1, blue sentences [R1-A2] and [R1-C10].

Table 2; SI Sec. 1.4 and Fig. S2a,b.

Table 2: **Multi-arm grouped controls and train–path overlap audit.** Test errors are mean ± sample SD across training seeds. G1 errors are fixed-image diagnostics evaluated against the legacy baseline. The 25-image overlap distances are separate from the current pointwise test metrics.

Grouped route comparison

Route	Train configs (seeds)	Test configs/seed	G1 $\lvert\Delta E\rvert$ (eV)	Test E RMSE (meV/atom)	Test F RMSE (meV/Å)
FT-600K	1,786 (3)	267	0.47 ± 0.12	41 ± 5	260 ± 6
Scratch-600K	1,786 (3)	267	1.20 ± 0.27	30 ± 3	344 ± 36
Scratch-5%	89 (3)	267	0.68 ± 0.20	17 ± 9	342 ± 15
FT-MultiT	2,659 (3)	767	—	33 ± 1	233 ± 2
Scratch-MultiT	2,659 (3)	767	—	30 ± 3	414 ± 192
MPA0 (zero-shot)	— (1)	267	0.24	39	279

Overlap audit over 25 legacy migration images

Training subset	Train frames	Exact matches[a]	Min. SOAP cosine	RMSD candidates	Min. Cr-local RMSD
Full grouped 600 K	1,786	0	5.4×10^{-4}	2	2.06 Å
Scratch-5% subset	89	0	6.4×10^{-4}	1	2.06 Å

[a]No exact global or Cr-local matches; the SOAP near-duplicate cutoff was 10^{-4}. RMSD near-duplicate cutoff: 0.05 Å.

A3. Uncertainty in kinetic results

> *"Add uncertainty to the kinetic results: retrain at least Scratch and FT-600K with ≥3 seeds and report barrier-error mean +/- std (the 3-seed protocol already exists for Table S1). This simultaneously tests the 'chance' explanation for the scratch model's deep-penetration result—if the low error is seed-stable, 'chance' is refuted; if seed-dependent, it is supported. Either way, replace the current unfalsifiable assertion with evidence."*

Response. To address this, we now report evaluations across three independent training seeds for every contemporary trained condition in the main task matrix. For the canonical fixed within-QL images, Scratch-600K yields a mean absolute reference discrepancy of 0.45 ± 0.05 eV. Because this discrepancy remains stable across initializations, the data indicate that the relatively low error is a reproducible characteristic of the scratch training route, rather than a statistical artifact or chance occurrence.

The legacy control pathways exhibit varying degrees of seed dependence. Pathway Q1 displays a stable Scratch discrepancy of 0.51 ± 0.11 eV, whereas Q2 and Q3 show higher variance (0.72 ± 0.55 eV and 1.77 ± 1.00 eV, respectively). All reported uncertainties are sample standard deviations over training seeds, which explicitly quantify model initialization variance.

Table 3: **Seed-resolved fixed-image errors by legacy pathway class.** Entries report the mean ± sample standard deviation over three independent training seeds. G1, G2, and Q1–Q3 serve as baseline image-set diagnostics rather than macroscopic dynamic migration barriers.

Pathway class	Model	$\lvert\Delta E_{\mathrm{peak}}\rvert$ (eV)
Q1 (gap-to-QL)	Scratch	0.51 ± 0.11
	FT-600K	9.60 ± 0.57
Q2 (gap-to-QL)	Scratch	0.72 ± 0.55
	FT-600K	10.40 ± 0.44
Q3 (gap-to-QL)	Scratch	1.77 ± 1.00
	FT-600K	9.70 ± 0.20
G1 (within-gap)	Scratch	1.20 ± 0.27
	FT-600K	0.47 ± 0.12
	Scratch-5%	0.68 ± 0.20
G2 (within-gap)	Scratch	0.33 ± 0.06
	FT-600K	0.21 ± 0.18
Current canonical reference (separate image set)		
Within-QL	Scratch-600K	0.45 ± 0.05

Takeaway. Seed-resolved evaluations confirm that the fixed-image energy assignments of the scratch model are reproducible. However, because these same scratch conditions fail rapidly in MD, the revised text emphasizes that stable static energy assignments do not necessarily translate to reliable dynamic path generation. These legacy pathway evaluations are retained strictly as static diagnostic metrics rather than dynamic kinetic predictions.

Where to find the change. Main Sec. 2.3, Fig. 5c and Fig. 3a–f.

SI Tables S4 and S7; Sec. 2.2.

A4. Scope of generalizability claims

> *"Rescope the generalizability claims: change 'establishes ... a generalizable standard' to a candidate framework demonstrated on one system, and move 'broader principles' to a hedged outlook—or add one additional material/dopant (even a cheap second system, e.g., a different transition-metal dopant in the same host) as minimal cross-validation."*

Response. We agree that precisely bounding the scope of the claims strengthens the scientific rigor of the manuscript. Accordingly, the title, abstract, and Discussion have been revised to strictly contextualize the findings within the process-specific reliability of the tested MACE implementations in Cr-doped Sb_2Te_3. The manuscript now presents this workflow as a methodological blueprint demonstrated on a primary system, rather than asserting a universal model-selection standard or a quantitative scaling rule based on foundation pretraining distance.

The supplementary Cu/Li/Na pathway profiles and the six-system endpoint-perturbation evaluations are retained as targeted qualitative diagnostics. These specific tests evaluate localized structural perturbation responses; therefore, they are appropriately scoped and are not extrapolated into macroscopic transferability claims across new materials.

Table 4: **Scope of the evidence across systems.** Additional materials systems serve as targeted diagnostics with defined evaluation boundaries.

Evidence set	Coverage	What it supports
Cr-doped Sb_2Te_3	Full grouped, path, and MD comparisons	Primary-system conclusions
Cu/Li/Na in Sb_2Te_3	Supplied fixed-path profiles	Qualitative pathway diagnostics
Six dopant–host systems	846 endpoint relaxations	Local structural perturbation response

Takeaway. The conclusions are now explicitly bounded to the primary Cr-doped Sb_2Te_3 system. The evaluation framework is positioned as a methodological blueprint, with broader cross-system validation designated as future work.

Where to find the change. Main Introduction and final paragraph of Sec. 3, marked [R1-A4].

Table 4; SI Fig. S3a–l and Sec. 7.

A5. Scratch-5% control

> *"Add a Scratch-5% control (same ∼1,000 configurations as FT-600K, random initialization) to disentangle foundation-pretraining benefits from data-volume effects."*

Response. To address this, the revised primary 600 K evaluation now trains both the full scratch and fine-tuned models on the identical 1,786-frame dataset. Both routes utilize the exact same 311/267 validation/test partitions and are evaluated across three independent seeds. This fully matched protocol eliminates target-data volume as a confounding variable in the primary comparison. Because architecture and optimization inherently differ between these registered routes, this matched-data evaluation assesses their integrated macroscopic effects rather than isolating pretraining alone.

With the primary comparison now fully data-matched, the separate Scratch-5% run (which comprises 89 configurations) is explicitly defined in the revised manuscript as a data-volume sensitivity baseline, rather than a control for the grouped FT model. Its G1 discrepancy of 0.68 ± 0.20 eV is retained solely to illustrate baseline model behavior under conditions of extreme data scarcity.

Takeaway. The primary evaluation now utilizes an identical 1,786-frame budget for both scratch and fine-tuned routes to ensure a direct, volume-controlled comparison. The separate 89-frame run is retained exclusively as a low-data sensitivity baseline.

Where to find the change. Main Sec. 5.1, blue sentence [R1-A5].

Table 2; SI Secs. 1.2 and 1.4.

A6. Interpretability claims: t-SNE/PHATE and SHAP

> *"Soften the interpretability claims: reframe t-SNE/PHATE conclusions as 'consistent with' rather than 'direct mechanistic explanation,' report projection sensitivity to perplexity/seed, and compute at least one separation metric in the original descriptor space. For SHAP, report per-model surrogate R^2, rescope 'the model internalizes X' to 'the surrogate's error predictions are most sensitive to X,' and (ideally) add a perturbation test on MACE's own inputs for the dominant Cr-Cr channel."*

Response. We agree that bounding the interpretability claims strengthens the physical grounding of the analysis. Accordingly, we have refined the trajectory projections to serve as macroscopic structural diagnostics. The revised text clarifies that these projections map explicit output features (recorded energies, forces, and structural summaries) rather than internal MACE hidden states. To establish a baseline, we now report original-space separation metrics alongside projection sensitivity. For the current 927-frame matrix, the raw original-space silhouette is 6.3×10^{-2} (standardized: -5.9×10^{-2}), while the corresponding t-SNE score is 1.2×10^{-1}. We contextualize these layouts against energy offsets, feature scaling, and unequal trajectory horizons.

For the SHAP analysis, the revised manuscript frames the results as surrogate predictions of absolute energy error per atom on a common 267-frame trajectory. We now report the five-fold diagnostic R^2 values, which confirm surrogate fidelity (e.g., $0.98 \pm 3 \times 10^{-3}$ for FT-600K). These metrics quantify the in-distribution fidelity of the surrogate across training seeds, mapping local feature associations without overextending to out-of-distribution generalization.

Furthermore, the seed-resolved SHAP rankings reveal variance across identical architectures. This indicates that models capture pathway behavior through distributed, degenerate feature representations rather than a single dominant Cr–Cr causal channel. The supplementary coordinate-perturbation study (comprising 846 native-MACE endpoint relaxations over six dopant–host systems) evaluates local responses to explicit atomic displacements. This evaluation confirms

Table 5: **Multi-arm SHAP surrogate fidelity.** Five-fold diagnostic R^2 values are reported as mean ± sample standard deviation over three training seeds on the same 267-frame test trajectory. These metrics quantify surrogate-fit fidelity rather than macroscopic MACE validation scores.

Model family	Seeds	Five-fold R^2
FT-600K	123, 234, 345	$0.98 \pm 3 \times 10^{-3}$
FT-MultiT	123, 234, 345	$0.97 \pm 3 \times 10^{-3}$
Scratch-600K	123, 234, 345	$0.98 \pm 6 \times 10^{-3}$
Scratch-MultiT	123, 234, 345	$0.95 \pm 2.2 \times 10^{-2}$

physical-input robustness and supports the conclusion that macroscopic errors arise from complex, multi-channel structural interactions acting collectively.

Takeaway. The interpretability claims are bounded. Trajectory projections serve as macroscopic population diagnostics, and SHAP analyses quantify in-distribution error-surrogate associations, ensuring the structural metrics are not overstated as direct mechanistic causation.

Where to find the change. Main Secs. 2.5–2.6; Fig. 6a–d and Fig. 7a,b.

Tables 5 and 9; SI Secs. 5–6 and endpoint appendix.

A7. Related work and distinction from our earlier report

> *"Position against the nearest neighbors: cite and discuss arXiv:2510.05020 (MACE fine-tuning for Li-ion diffusion), the foundational-MLIP migration-barrier evaluation (RSC Digital Discovery d5dd00534e), and arXiv:2405.07105 (systematic softening—currently invoked as if novel). Add one paragraph explicitly stating this manuscript's delta over the authors' own arXiv:2509.00090 [46]."*

Response. We have responded to the reviewer's request to recommend relevant studies that will help delineate our work relative to prior literature and our earlier preprint.

The revised Introduction now incorporates and discusses all three related studies. We highlight the work of Alghamdi *et al.* on zero-shot and adapted MACE diffusion in LiF, the work of Bheemaguli *et al.* separating migration-energy and path-geometry metrics, and the findings of Deng *et al.* regarding systematic softening [1–3]. The observed within-QL energy overestimates are now framed within the context of systematic softening established by Deng *et al.*, presenting these results as an empirical observation within doped topological chalcogenides rather than a newly identified physical phenomenon.

We have also clarified the delta of this manuscript relative to our earlier non-archival preprint [4]. While that preliminary report explored initial migration, sliding, and projection behaviors, the present manuscript provides a controlled evaluation framework. Specifically, the current study adds:

1. Strict source-trajectory partitioning to prevent adjacent-frame data leakage;
2. Three-seed evaluations across all trained conditions to quantify model initialization variance;
3. Explicit decoupling of static fixed-image calibration from model-driven path generation;
4. Quantitative failure-horizon tracking in molecular dynamics trajectories;
5. In-distribution SHAP error-surrogate analysis and original-space structural distance checks.

Furthermore, the legacy single-run 0.16 eV and 0.41 eV values have been replaced by the standardized, multi-seed benchmarks reported in the revised main text.

Takeaway. The Introduction now contextualizes the findings within existing literature, properly attributing prior work on systematic softening. Relative to our earlier preprint, this manuscript establishes a multi-seed, trajectory-partitioned benchmark that decouples static accuracy from dynamic stability.

Where to find the change. Main Sec. 1, related-work and study-scope paragraphs; marker [R1-A7]. The non-archival report distinction is addressed in this response because the reviewer specifically requested it.

The corresponding controls and corrections are documented in SI Secs. 1.4, 2.2, 5 and 6.

C8. Zero-force NEB exporters

> *"Fake zero forces in NEB→training-data exporters (`examples/Sb2Te3_Cr_doped/03_neb/neb_data_collector.py:442-443`, `neb_to_extended_xyz.py:278-282`): real QE energies paired with fabricated all-zero forces written into training-format extxyz under `forces:R:3`; identical bug in two scripts. If any such frames fed fine-tuning (`forces_weight=100` per repo README), the model was silently trained toward zero forces. Not proven wired into FT-600K/FT-MultiT (merge config references nonexistent folders). → Export real forces from QE output or hard-disable; authors must confirm/deny use in reported runs."*

Response. We have updated both affected scripts (`neb_data_collector.py` and `neb_to_extended_xyz.py`) to enforce fail-closed behavior whenever first-principles forces are unavailable. The collector now blocks any force-bearing export unless validated electronic-structure forces are present. Similarly, the extended-XYZ exporter requires valid forces by default and raises an explicit error if called without validated forces, writing energy and geometry only when explicitly flagged with `--no-forces`. Fabricated zero-force arrays are completely disabled.

To verify whether any unphysical zero-force frames entered the training pipelines, we performed a provenance audit across all 9,163 records in the declared historical and grouped training splits. As summarized in Table 6, exactly nine records contain identically zero forces, all corresponding to documented single-atom Cr, Sb, and Te isolated-atom energy references. The audit confirmed zero unclassified and zero NEB-derived zero-force frames across the training splits. The frame manifest, machine-readable summary, audit script, and corrected exporters are included in the revision package.

Table 6: **Force-provenance audit across declared training-data files.** Inspected training records confirm the absence of unclassified or NEB-derived zero-force configurations.

Audit metric	Records
Total records inspected	9,163
Records with identically zero forces	9
Zero-force isolated-atom energy references	9
Unclassified zero-force records	0
NEB-derived zero-force records	0

Takeaway. Exporter scripts are now secured with fail-closed checks against missing forces. A provenance audit of all 9,163 training records confirms that no fabricated zero-force frames entered the training lineage of any reported model.

Where to find the change. Main Sec. 5.1, blue sentence [R1-C8].

SI Sec. 1.3; accompanying zero-force manifest and corrected exporters.

C9. Matched MD protocol for FT-600K

> *"Asymmetric MD protocol for the naive-FT condition (`results/MD/2-naive-fine-tuning/mace_md_eval_0814.py:359-360: steps=2000, interval=5` vs. `steps=100000, interval=200` in the other three byte-identical scripts): 50× shorter trajectory with 40× denser sampling in the cross-model transport comparison; post-processing assumes 200 fs/frame, so D/κ for that condition may be off ∼40×."*

Response. To address this, we have standardized the MD continuation protocol across all evaluated conditions. Every run now utilizes an identical 1 fs timestep and a 200-step output interval. All trajectories evaluate direct continuations from a shared 200 ps state toward a full 1,000 ps (1 ns) target clock.

Under this uniform evaluation, the macroscopic stability limits of the models become readily apparent. The three foundation checkpoints and FT-600K seed 234 successfully completed the full 1 ns target trajectory. The remaining fine-tuned and scratch models terminated at distinct physical failure horizons. We now report these specific failure horizons directly, replacing the prior extrapolations of long-time transport coefficients derived from unstable runs. In this evaluation, the training seeds capture model initialization variance, while the shared initial momenta evaluate the deterministic dynamic response.

Table 7: **Standardized MD continuation endpoints.** Final simulation clocks are reported as recorded under a uniform integration protocol. Stable models successfully completed the 1 ns target without structural collapse.

Model family	Seed / checkpoint	Final clock (ps)	Continuation (ps)	Outcome
Foundation	MPA-0	1,000	800	Target reached
	MP-0b3	1,000	800	Target reached
	OMAT-0	1,000	800	Target reached
FT-600K	123	236	36	Scientific stop
	234	1,000	800	Target reached
	345	485	285	Scientific stop
FT-MultiT	123	237.2	37.2	Scientific stop
	234	270.2	70.2	Scientific stop
	345	286	86	Scientific stop
Scratch-600K	123	204.4	4.4	Scientific stop
	234	200.8	0.8	Scientific stop
	345	200.4	0.4	Scientific stop
Scratch-MultiT	123	201.2	1.2	Scientific stop
	234	200.4	0.4	Scientific stop
	345	200.4	0.4	Scientific stop

Takeaway. The MD analysis now evaluates structural stability under a uniform 1 ns integration protocol. Macroscopic transport extrapolations are removed to ensure all physical claims remain strictly bounded by the observed trajectory horizons.

Where to find the change. Main Sec. 2.2, Fig. 3a–f.

Table 7; SI Sec. 4 and Table S8.

C10. Trajectory-grouped train/test split

> *"Ungrouped random train/test split (`split_data.py:52-66`): frame-level shuffle over merged trajectories with no grouping by pathway/trajectory ID; near-duplicate adjacent frames can land in both train and test, inflating Table S1 test RMSE. Provenance metadata embedded by the collector (`path_name`, `neb_image`) is never read downstream—no NEB-exclusion safeguard exists."*

Response. To address this, we have updated the data partitioning pipeline. The revised protocol now actively utilizes the provenance metadata (`path_name`, `neb_image`) to enforce trajectory-level grouping and strictly exclude NEB-derived frames from the training split. Complete source trajectories are now assigned atomically to either the train, validation, or test sets. This design eliminates the temporal leakage associated with the previous frame-level shuffle, ensuring the test metrics evaluate independent generalization.

The matched 600 K evaluation utilizes this grouped 1,786/311/267 (train/validation/test) configuration split. The revised main parity figure reports the grouped training and held-out test data separately, and the caption distinguishes unique configurations from the pooled prediction count. Full uncentred test errors and corresponding data SHA256 hashes are provided to ensure reproducibility.

Table 8: **Trajectory-grouped 600 K dataset partitions.** The grouped split ensures temporal isolation. Hashes are abbreviated to their first 12 hexadecimal characters for reference.

Split	Configurations	Data SHA256 prefix
Train	1,786	`04b1369d0da0`
Validation	311	`aa9a9502854d`
Test	267	`d49cf4b2ab49`

Takeaway. Temporal data leakage is eliminated through trajectory-level grouping, driven directly by the provenance metadata. The revised test partitions provide a temporally isolated measure of out-of-distribution generalization.

Where to find the change. Main Secs. 2.1 and 5.1, marker [R1-C10]; Fig. 2a–h.

Tables 2 and 8; SI Secs. 1.2 and 1.4.

C11. Silhouette scores in the original descriptor space

> *"Silhouette scores computed on the 2D t-SNE embedding, not the original feature space (`results/t-SNE/3d/tsne.py:323`, `phate-tsne/tsne.py:191`, `mace_probe_blackbox0818.py:254`): the Fig. 5d separability numbers measure projection artifacts, not true latent separability. Need to compute in original/high-dimensional space."*

Response. To address this, we have computed the separation diagnostics directly within the original high-dimensional output feature space across the complete 927-frame dataset. As detailed in Table 9, the original-space Euclidean silhouette is 6.3×10^{-2}, the cosine silhouette is 6.1×10^{-2}, and the standardized Euclidean silhouette is -5.9×10^{-2}. A complementary 30-bin neighbour histogram yields 5.6×10^{-2}. These direct, high-dimensional metrics now serve as the primary quantitative basis for our feature analysis.

The manuscript now explicitly pairs these original-space metrics with the visual embeddings. Panel d of the main projection figure continues to report the t-SNE score to quantify the visual layout, but the text and caption now explicitly designate it as a 2D projection metric, decoupling it from the original-space separability analysis. Furthermore, the Supplementary Information provides a

t-SNE sensitivity check across three perplexities (5, 30, and 50). Under PCA initialization, the independent seeds produce identical embeddings, confirming stable topological behavior.

Table 9: **High-dimensional and projection silhouette diagnostics.** The 927-frame dataset evaluations separate original-space structural metrics from 2D projection layouts. Under PCA initialization, the three seeds at each perplexity produced identical embeddings.

Space or projection	Dimensions / setting	Silhouette	Trustworthiness
Original output features, Euclidean	6,191	6.3×10^{-2}	—
Original output features, cosine	6,191	6.1×10^{-2}	—
Z-scored output features	6,191	-5.9×10^{-2}	—
Common neighbour histogram	30	5.6×10^{-2}	—
Z-scored neighbour histogram	30	5.2×10^{-2}	—
t-SNE, PCA init.	perplexity 5	7.9×10^{-2}	0.97
t-SNE, PCA init.	perplexity 30	1.2×10^{-1}	0.98
t-SNE, PCA init.	perplexity 50	1.3×10^{-1}	0.97

Takeaway. The interpretation of the feature space is now grounded directly in original-space separation metrics. The consistently low high-dimensional silhouette scores confirm the presence of highly degenerate feature representations, establishing that the macroscopic model behaviors are not driven by cleanly separated latent states.

Where to find the change. Main Sec. 2.5, and Fig. 6d caption.

SI Table S9, Table S10 and Fig. S5a,b.

C12. Inactive force-sensitivity feature stub

> *"Always-zero 'force sensitivity' feature stub in all three latent-probe scripts (`3d/tsne.py: 209-215`, `phate-tsne/tsne.py: 132`, `mace_probe_blackbox0818.py: 171-180`), silently concatenated into the feature matrix feeding PCA→t-SNE→PHATE."*

Response. We thank the reviewer for their meticulous code review, identifying this inactive feature stub. We agree that the feature matrix should be explicitly audited to contain only active, information-bearing dimensions.

To address this, we have updated the data processing pipeline to systematically audit and remove any constant columns prior to dimensionality reduction. As verified by the current executable audit, the identically zero placeholder (the final column) is now explicitly stripped before any distances or projections are computed, leaving a dense matrix of 6,191 active features.

Mathematically, a constant zero coordinate contributes zero variance and has identically zero effect on Euclidean or cosine distance calculations. Therefore, while this placeholder existed in the legacy feature grammar, it exerted no mathematical influence on the resulting topological layouts or structural embeddings. Explicitly removing it now ensures the quantitative workflow is fully transparent and properly conditioned. Furthermore, the separate SHAP analysis utilizes an independent matrix of computed SOAP and structural descriptors, which is entirely unaffected by this downstream visualization pipeline.

Takeaway. The data pipeline now explicitly audits and removes the inactive placeholder prior to analysis. Because a constant column has zero mathematical contribution to distance metrics, the quantitative evaluations and structural embeddings are driven solely by the 6,191 active physical features.

Where to find the change. Main Sec. 5.3, blue sentence [R1-C12]; Fig. 6a–c caption.

SI Sec. 5; executable current-feature audit.

Table 10: **Feature-matrix audit.** The pipeline now explicitly verifies the removal of the constant placeholder prior to distance and projection computations.

Audit metric	Result
Frames	927
Input feature columns	6,192
Zero-valued placeholder	final column (index 6,191; zero-based)
Placeholder values equal to zero	all 927 frames
Columns retained after removal	6,191

C13. SHAP code and repository completeness

> *"SHAP analysis code absent from the repository entirely; advertised* `migrationbench` *package (36 files),* `scripts/` *(13 files including* `shap_analysis.py`*), templates, workflow docs, and* `paper/manuscript.tex` *are all 0-byte stubs. Section 3.5 results and the 'benchmark framework' API are not reproducible from this release."*

Response. To address this, the revised submission package now provides the fully populated, executable SHAP analysis workflow and data infrastructure. The accompanying repository contains the common 267-frame XYZ trajectory, all 12 aligned evaluation tables, feature-label mappings, per-seed metrics, and the rendering scripts for the feature attributions. The current-feature audit and its input identities are supplied concurrently.

We have independently verified the numerical reproducibility of this workflow in the pinned environment. As detailed in Table 11, re-executing the supplied analysis from the source tables produces the 12 per-seed metric rows, 1,628 family-feature rows, 4,884 seed-feature rows, and 3,204 frame-aligned rows with zero numerical deviation from the reported figures.

Table 11: **Numerical reproduction check for the supplied SHAP outputs.** Executing the provided workflow in the pinned environment yields zero numerical deviation from the source tables.

Output table	Rows	Columns	Maximum numerical difference
Per-seed metrics	12	11	0
Family-feature aggregate	1,628	9	0
Seed-feature importance	4,884	6	0
Frame-aligned attributions	3,204	10	0

The current common-frame, four-condition analysis replaces the preliminary 150-frame reconstruction. The Methods and SI detail the 407 features, the estimator configurations, and the interpretation scope of the shuffled-fold validation. The quantitative claims are now tied directly to these validated workflows and figure-specific source tables, replacing the preliminary API structure from the initial snapshot.

Takeaway. The revision provides a complete, executable workflow. Numerical reproducibility is verified through provided figure-specific source tables, ensuring all reported metrics and attributions can be exactingly reconstructed.

Where to find the change. Main Code availability, marker [R1-C13], and Sec. 5.3.

Tables 5 and 11; SI Sec. 6; accompanying executable source and figure data.

Concluding Remarks

We thank the Editor and the Reviewers for their time, expertise, and constructive feedback. The detailed review process has fundamentally strengthened this manuscript. By addressing the raised methodological and physical concerns, we have transitioned the work from a preliminary exploration into a highly controlled, reproducible benchmark for evaluating foundation model reliability in active materials discovery.

We believe the revised manuscript now fully meets the stringent scientific standards of the journal. The updated evaluation framework not only clarifies the orthogonal nature of static accuracy and dynamic stability, but also provides a clear methodological blueprint for the materials modeling community. We look forward to your decision.

Sincerely,

Yi Cao, Peter Mastracco, Yile Wu, Victor Wu, Joshitha Pilla, and Paulette Clancy

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When Does a Foundation Model Help? Process-Specific Reliability of Machine-Learned Interatomic Potentials

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Abstract

Universal machine-learned interatomic potentials (MLIPs) are emerging as foundation models for materials science, promising out-of-the-box atomistic simulations across diverse chemistries. Their practical value, however, depends on whether they reliably reproduce the specific physical processes required for an application. Using Cr-doped Sb_2Te_3 as a representative layered system featuring distinct interfacial and bulk environments, we compare specialist, zero-shot, and fine-tuned MLIPs across thermal configurations, dopant migration, and collective layer sliding. Fixed-geometry energies, model-relaxed pathways, and molecular dynamics expose complementary failure modes. Target adaptation improves force accuracy and selected migration energies, yet fails to guarantee stable dynamics or reliable energies under strong distortions. Conversely, zero-shot checkpoints sustain longer trajectories while exhibiting calibration errors on specific migration configurations. Path relaxation can lower apparent barriers without recovering reference energy profiles. Structural projections and error surrogates characterize these differences without isolating their microscopic causes. These results establish a mandate for process-specific model selection: a useful potential must be tested both on configurations relevant to an observable and on those generated by its own forces. Foundation initialization is thus a resource to validate and adapt, rather than a guarantee of dynamical and thermodynamic reliability.

Keywords: machine-learned interatomic potentials (MLIPs), foundation models, fine-tuning, migration, distribution shift, model validation

1 Introduction

The advent of universal machine-learned interatomic potentials (MLIPs) is fundamentally shifting the paradigm of atomistic simulation. Rather than parameterizing a specialist potential for every new material, researchers can now initialize from a model trained across vast chemical spaces and either deploy it zero-shot or adapt it using a targeted reference dataset. MLIPs have made this prospect highly attractive by enabling first-principles accuracy at length and time scales inaccessible to density functional theory (DFT) alone [1–14]. Broadly trained universal potentials such as MACE-MP and CHGNet explicitly extend this capability across the periodic table [15–21]. The resulting choice is increasingly practical: should one deploy a universal checkpoint zero-shot, fine-tune it, or train a specialist from scratch?

For materials applications, the answer must ultimately be evaluated through the lens of specific physical processes. A model may reliably reproduce typical thermal fluctuations yet entirely misrepresent the short-lived coordination changes governing dopant motion. It may remain numerically stable while exploring a fundamentally inaccurate energy landscape. Conversely, minimal point-wise energy errors on a supplied transition path do not guarantee that the model's own generated forces will recover that pathway during a simulation. These distinctions are critical because activated motion is dictated by relative energies along rare transition configurations, whereas molecular dynamics (MD) autoregressively relies on the model's own force predictions to dictate subsequent phase-space exploration. Average error over a static test set and dynamical reliability in a target process are therefore fundamentally distinct metrics [22–26]. While broad materials benchmarks such as Matbench and JARVIS-Leaderboard assess property-prediction performance at scale, and generative benchmarks evaluate structural novelty, neither framework directly probes dynamical fidelity in complex, out-of-equilibrium physical processes [27–29].

Recent work establishes both the promise and the conditional nature of transfer. Alghamdi *et al.* show that zero-shot and lightly fine-tuned MACE models can reproduce Li diffusion observables in LiF; Bheemaguli *et al.* distinguish migration-energy accuracy from transition-path geometry across battery materials; and Deng *et al.* identify systematic softening of universal potentials on several out-of-distribution tasks [30–32].[R1-A7] These studies make process-aware testing essential and indicate that adaptation success depends heavily on which regions of the energy landscape are constrained by target data [33–39]. Yet, a critical question remains: how model performance shifts when evaluating the same material across fundamentally distinct physical processes and testing protocols.

Addressing this question requires an ideal model host that exposes a single chemical system to diverse, highly non-equivalent physical processes. We select Cr-doped Sb_2Te_3 as a representative benchmark system because its layered architecture natively

decouples compositional complexity from multi-regime distribution shifts. Its quintuple layers comprise a Te–Sb–Te–Sb–Te stack separated by van der Waals gaps. A Cr atom migrating within the gap, approaching a layer boundary, or entering the interior encounters diverse coordination environments, whereas layer translation induces a collective registry change. This combination provides a compact set of physically distinct probes within a single host (Fig. 1a,c), decoupling process-specific distribution shifts from compositional changes. More broadly, layered 2D materials are widely investigated across spintronic, quantum, and energy applications, where stacking, intercalation, and magnetic doping serve as primary tuning levers [40–55]. Cr-doped Sb_2Te_3 represents an established platform for exploring the interplay between transition-metal doping and layered topological matter [56–67].

Our comparative framework isolates two orthogonal dimensions of complexity. The first involves the nature of the evaluated configurations, transitioning from thermal fluctuations to dopant migration and collective sliding. The second involves how these configurations are generated: static reference paths test fixed energy assignments, whereas MLIP-relaxed bands and MD trajectories probe the feedback loop between model predictions and subsequent phase-space evolution. This distinction clarifies the specific benefits of foundation initialization and target adaptation, while revealing where apparent improvements fail to transfer.

Here we systematically evaluate how foundation initialization, target-data adaptation, and specialist training perform across thermal, migrational, and sliding regimes within a single host system. By directly connecting static held-out accuracy with model-generated dynamics, we map process-specific strengths and failure modes across these distinct physical regimes. Rather than pursuing a single global ranking, this work establishes a process-aware validation protocol for selecting, fine-tuning, and deploying interatomic potentials based on target physical observables.

2 Results

To determine how training lineage and initialization dictate model reliability, we evaluate specialist, zero-shot, and fine-tuned potentials across a hierarchy of physical processes in Cr-doped Sb_2Te_3. We begin by examining pointwise energy and force accuracy on held-out test data, and subsequently test how these static metrics translate into dynamical stability, reaction-path fidelity, and collective deformation under out-of-equilibrium conditions.

2.1 Pointwise accuracy gives ambiguous signals for energies and forces

Held-out evaluations show that energy and force metrics yield conflicting model rankings, highlighting the inadequacy of static scalar errors for predicting functional model suitability (Fig. 2a–h). Fine-tuning consistently lowers test force errors relative to scratch training across both target-data regimes (Fig. 2e–h), whereas energy errors follow a distinct, non-monotonic trend (Fig. 2a–d). On the 600 K test partition, FT-600K achieves a lower force RMSE (260 ± 6 meV·Å^{-1}) than Scratch-600K

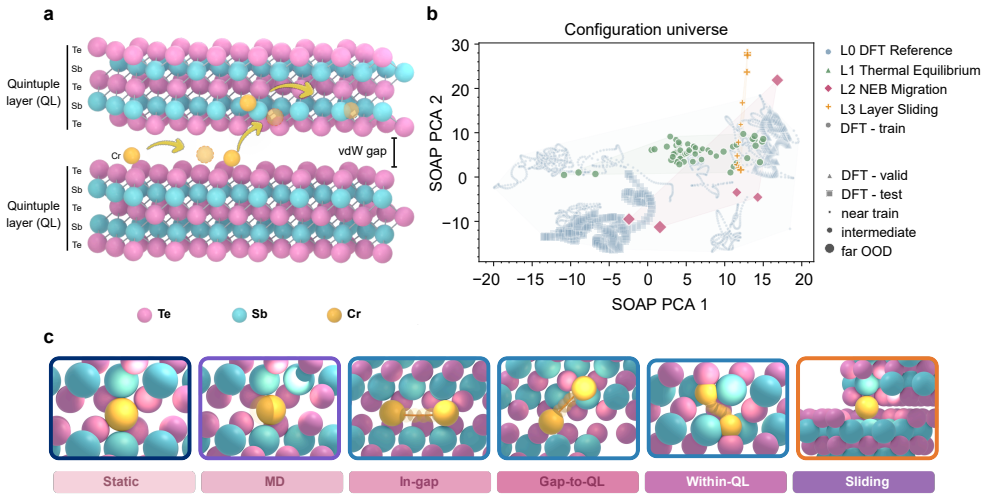

Fig. 1 A single host system encompasses diverse evaluative regimes for interatomic potentials. a, Cr-doped Sb_2Te_3, highlighting the Te–Sb–Te–Sb–Te quintuple layers (QLs), van der Waals gaps, and representative dopant displacements (Te, Sb, and Cr shown in pink, cyan, and yellow, respectively). **b,** SOAP-based principal component projection mapping reference configurations, thermal trajectories, migration images, and layer-sliding states. Symbols denote configuration sources and reference split assignments; this projection provides a descriptive visualization of local descriptor coverage rather than an absolute metric of extrapolation risk. **c,** Local atomic environments corresponding to static configurations, molecular dynamics, in-gap migration, gap-to-QL motion, within-QL motion, and interlayer sliding. These task categories organize the physical probes; their sequential layout reflects distinct operational regimes rather than a monotonic distribution-shift scale.

(344 ± 36 meV·Å^{-1}), yet exhibits a higher energy RMSE (41 ± 5 vs. 30 ± 3 meV·atom^{-1}; Supplementary Table S1).

These results illustrate that optimizing force predictions does not guarantee a corresponding improvement in the energy landscape, nor does a lower static energy error ensure better force accuracy. Because these pointwise metrics offer ambiguous guidance on overall potential quality, process-aware evaluations across dynamical and out-of-equilibrium tasks become essential to determine downstream physical reliability.

2.2 Trajectory survival does not guarantee local dynamical fidelity

Trajectory survival differs sharply across the tested training routes (Fig. 3a–f). All three zero-shot foundation checkpoints successfully reach the 1 ns target. Among the adapted checkpoints, only FT-600K (seed 234) achieves this duration, whereas the remaining two FT-600K runs terminate at 236 and 485 ps, and FT-MultiT runs end between 237.2 and 286 ps. All six scratch models collapse at a scientific stopping threshold between 200.4 and 204.4 ps (Supplementary Table S8). These trajectories demonstrate that zero-shot foundation checkpoints systematically sustain longer usable simulation horizons than adapted or specialist counterparts.

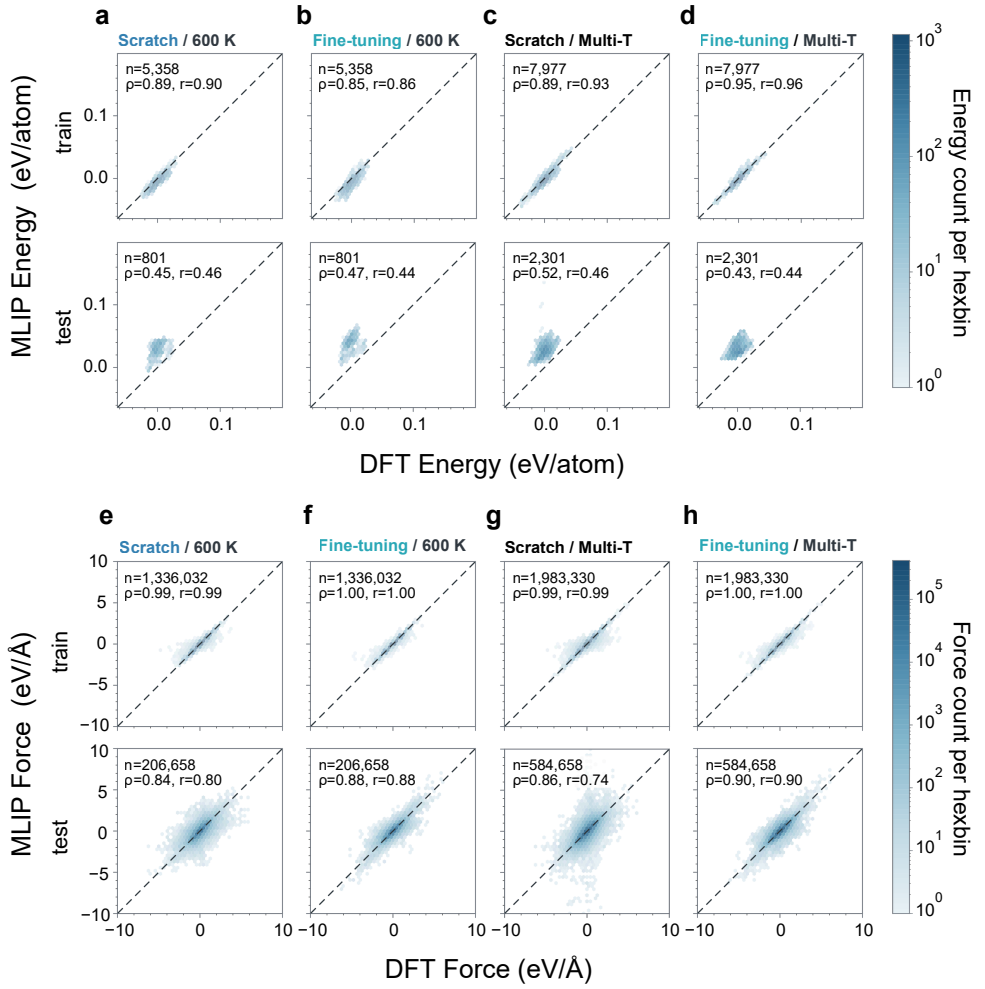

Fig. 2 Pointwise evaluation of energy and force agreement reveals conflicting model rankings across training lineages. a–d, Energy parity plots for Scratch-600K, FT-600K, Scratch-MultiT, and FT-MultiT models, respectively; each panel displays training data (upper) and held-out test data (lower). **e–h,** Corresponding Cartesian force-component parity plots in the same model sequence. Energies are expressed per atom and are relative to the pooled DFT mean of each trajectory cluster (test-only cluster means are applied solely for visualization). Forces represent uncentred Cartesian components. Predictions from three independent training seeds are pooled, yielding n prediction records rather than independent structures: 5,358 training and 801 test energy records per 600 K family, and 7,977 training and 2,301 test energy records per MultiT family. Hexbin color density follows a logarithmic scale, with independent normalizations for energy and force. Dashed diagonal lines indicate ideal parity; ρ and r denote Spearman and Pearson correlation coefficients, respectively. The force display window is restricted to ± 10 eV·Å^{-1}, omitting 106 outlying Scratch-MultiT test components that remain fully accounted for in all reported numerical metrics. Plotted energy correlations serve as relative within-cluster diagnostics, whereas tabulated uncentred, seed-resolved errors (Supplementary Table S1) preserve absolute energy and force offsets.

However, trajectory survival alone does not guarantee local dynamical accuracy. Despite exhibiting similar principal peaks in all-atom radial distribution functions (Fig. 3d,e), foundation trajectories display significantly higher in-plane mean-square displacement (MSD) mobility and distinct local Cr coordination environments (Fig. 3c). This divergence proves that global structural averages can readily obscure local dopant discrepancies. While kinetic temperature remains well-regulated across stable runs, hydrostatic pressure and other thermodynamic indicators exhibit pronounced model-dependent variations (Fig. 3a,b), highlighting that different training lineages sample non-equivalent physical ensembles even when trajectories persist.

Because premature trajectory truncation precludes standard long-time transport analysis, these comparative simulations are not interpreted as equilibrium diffusion coefficients. Instead, we analyze recorded MD trajectories strictly to benchmark dynamical stability, local coordination, and short-time mobility across model lineages.

2.3 Model-relaxed migration pathways expose barrier collapse despite static energy accuracy

While MD reveals divergent local mobilities, isolating the energetic origins of these discrepancies requires probing specific transition pathways. To this end, we design a hierarchy of Cr migration probes: (1) in-gap diffusion (low distortion), (2) gap-to-QL entry (interfacial crossing), and (3) within-QL motion (strong local distortion) to untangle how foundation, fine-tuned, and specialist models navigate distinct physical terrains (Fig. 4a–f).

Migration energetics depend on both the spatial regime and whether evaluations use fixed reference geometries or unconstrained model-relaxed pathways. On fixed reference configurations, foundation and fine-tuned checkpoints reproduce the qualitative shape of the in-gap diffusion reference more accurately than legacy scratch models (Fig. 4a; Fig. 5b). However, this apparent transferability breaks down in the dense layer interior. For the within-QL trajectory, foundation and fine-tuned potentials assign excessive energy penalties, often exceeding the DFT reference by an order of magnitude, to strongly distorted intermediate configurations (Fig. 4c; Fig. 5c). In contrast, Scratch-600K achieves substantially smaller fixed-image discrepancies of 0.45 ± 0.05 eV (Supplementary Table S4). While this localized pointwise accuracy is reproducible across scratch initializations, these same models rapidly collapse during MD simulation, reinforcing that accurate static energy assignments are insufficient to construct a reliable global force field.

The discrepancies emerge when structural relaxation is driven entirely by the models' own forces. Unconstrained pathway relaxation unmasks unphysical deviations across all model lineages. Several MLIP-relaxed profiles exhibit spurious deep intermediate minima or suppressed activation barriers, entirely missing the reference topology (Fig. 4d–f). Notably, most within-QL forward peaks collapse toward zero despite their massive fixed-image energy overestimations (Fig. 5d). This barrier collapse results directly from unguided force-matching driving the system into unphysical local basins or altering the transition endpoint registry.

By benchmarking against fully converged, first-principles NEB reference pathways, we expose the uncoupling of static energy prediction and force-driven structural

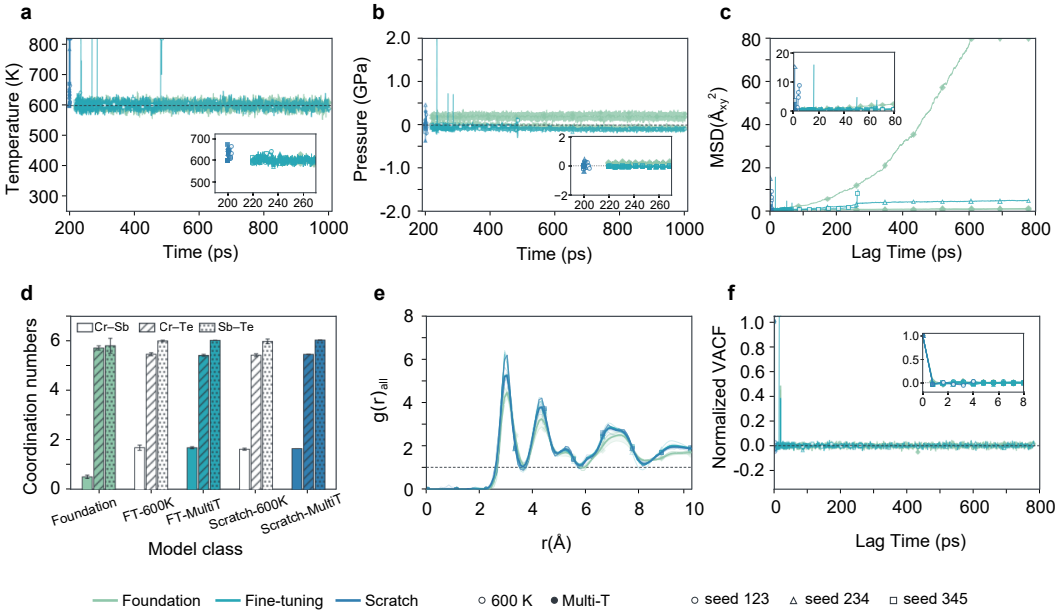

Fig. 3 Molecular dynamics simulations driven by self-generated forces reveal divergent stability limits and local structural evolution. a, Kinetic temperature profiles. **b,** Hydrostatic pressure, $P = -\mathrm{tr}(\sigma)/3$, computed from recorded stress tensors. **c,** In-plane, multiple-origin mean-square displacement (MSD) evaluated with triclinic periodic unwrapping. **d,** Pair-resolved coordination numbers determined using fixed Cr–Sb, Cr–Te, and Sb–Te cutoffs of 3.2, 4.1, and 4.2 Å, respectively (bars and whiskers denote the mean and sample standard deviation within each condition). **e,** All-atom radial distribution functions (RDFs) derived from ordered pair counts and triclinic minimum-image distances (thin traces represent individual models, and summaries incorporate display-only smoothing). **f,** Normalized multiple-origin velocity autocorrelation functions, with insets expanding early-time behavior. Color hue distinguishes model strategies, open versus filled symbols designate 600 K versus MultiT training regimes, and marker shapes identify training seeds (foundation variations reflect independent checkpoints). Simulations are initialized from a common configuration at 200 ps with a target duration of 1 ns (1 fs timestep, 200-step output interval). Trajectory curves terminate at their respective observed simulation horizons, and structural summaries integrate all available, unequal time windows. Only three foundation checkpoints and FT-600K seed 234 successfully sustain the full 1 ns target (Supplementary Table S8), indicating that trajectory survival represents a stringent test of model lineage that uncouples from static pointwise accuracy.

evolution. The inability of MLIPs to preserve validated activation barriers during self-relaxation underscores a methodological mandate: an optimizer-converged elastic band cannot support macroscopic kinetic interpretations without explicit structural and energetic validation against first-principles references.

2.4 Collective sliding tests what local adaptation leaves unconstrained

Relative layer displacement changes registry throughout the entire structure. It therefore probes a distinct response from moving a single Cr atom, even though both processes occur within the same host lattice. Figure 5e and Supplementary Fig. S4a–d

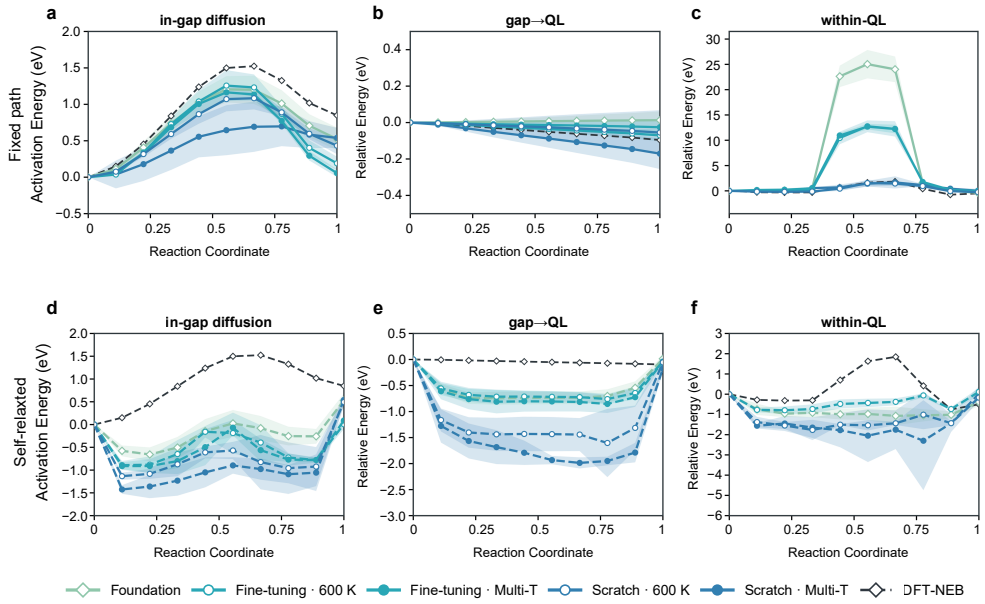

Fig. 4 Fixed-energy profiles and model-relaxed pathways expose process-dependent deviations across MLIP lineages. a–c, Energy profiles evaluated on fixed reference configurations for in-gap diffusion, gap-to-QL motion, and within-QL motion, respectively. **d–f,** Corresponding endpoint-initialized, MLIP-relaxed pathway diagnostics along the reaction coordinate. Energies are referenced to the initial image of each profile. Dashed dark curves denote the DFT reference data, while coloured curves and bands represent the mean and pointwise dispersion across selected model lineages. These comparisons demonstrate that while foundation and fine-tuned checkpoints reliably reproduce qualitative in-gap energetics on fixed geometries (**a**), they severely overestimate activation barriers under strong local distortions within quintuple layers (**c**). Furthermore, model-relaxed trajectories (**d–f**) reveal that unconstrained force-matching can artificially reduce apparent barriers or divert from true reference pathways, underscoring that accurate pointwise energy assignment does not inherently ensure physically faithful structural relaxation.

compare pristine and Cr-containing sliding profiles. Predicted energy peaks and profile shapes exhibit pronounced variations across the foundation, fine-tuned, and scratch model lineages. While specific foundation and fine-tuned checkpoints approach the correct reference peak magnitude, they frequently fail to reproduce the full topological profile or the nominal endpoint energy return.

This divergence emerges prominently in the fixed-path curves. The DFT sliding profiles return perfectly to their initial reference energy after a full periodic displacement, whereas MLIPs across different training routes retain an artificial endpoint offset. An accurate peak magnitude can therefore coexist with an incorrect structural registry response. This persistent endpoint offset demonstrates that learning localized coordination environments through either massive zero-shot pre-training or targeted fine-tuning fails to intrinsically constrain the global periodic energy registry.

By integrating these sliding responses with the migration and MD evaluations (Fig. 5), the combined results establish a strict decoupling of model capabilities. As demonstrated, a scratch training route achieving minimal static energy discrepancy

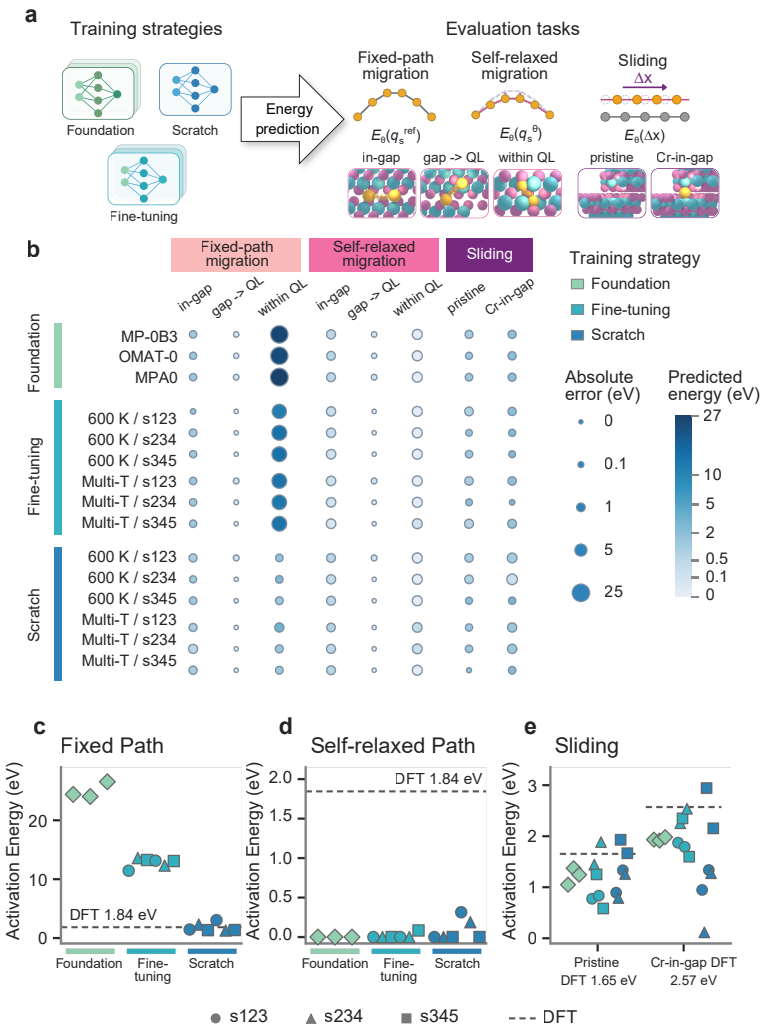

Fig. 5 Reliability depends on the task and evaluation operator. a, Evaluation design: three foundation checkpoints and three training seeds for each of the four trained conditions (scratch or fine-tuned, each using 600 K or MultiT data), evaluated over three fixed migration paths, three self-relaxed migration paths, and two fixed sliding paths. **b,** The 120-record source matrix. Colour encodes the predicted forward peak relative to the first image; circle area follows a square-root mapping of the absolute discrepancy from the reference. Colour uses a power normalization with exponent 0.40. A minimum dot remains at zero discrepancy. **c-d,** Within-QL forward peaks for fixed and self-relaxed paths; the reference line is 1.84 eV and the vertical scales differ. **e,** Fixed-path sliding peaks for pristine and Cr-in-gap systems, with reference peaks of 1.65 and 2.57 eV. Shapes identify training seeds or foundation checkpoints. All 120 validated pathway evaluations are retained as explicit diagnostic benchmarks. A minimal self-relaxed peak indicates local optimization convergence but does not intrinsically certify a physical transition barrier unless structural pathway correspondence is also verified. Fixed sliding endpoints and profiles are evaluated subject to exact thermodynamic consistency checks. Comprehensive numerical metrics accompany the figure.

collapses under its own MD. Conversely, a foundation checkpoint sustaining a stable MD trajectory severely misassigns local within-QL migration barriers. Finally, collective sliding imposes a periodic endpoint constraint that remains independent of these localized successes. Consequently, MD stability, local transition barriers, and collective sliding responses represent orthogonal dimensions of model reliability, dictating that no single performance metric can validate an MLIP for complex materials discovery.

2.5 Trajectory projections diagnose divergence within shared structural manifolds

The orthogonal failure modes observed across MD, local migration, and collective sliding mandate a direct interrogation of the models' representational mechanisms. To determine whether these divergences stem from exploring completely isolated structural phases or from differing internal feature sensitivities, we first map the extrinsic phase space of the simulations. Trajectory projections visualize the phase space explored by the 15 model conditions during the MD runs (Fig. 6a–c). These projected features incorporate recorded energies, forces, and structural summaries to capture the macroscopic evolution of each model. The resulting spatial separation visualizes distinct survival lifetimes and failure horizons, as the scratch trajectories terminate early and isolate their populations within the projected space.

Quantitative clustering metrics reveal the underlying structural relationship among these divergent trajectories. The unprojected output-feature silhouette is 6.3×10^{-2}, shifting to -5.9×10^{-2} after feature standardization, while the two-dimensional t-SNE projection yields 1.2×10^{-1} (Fig. 6d; Supplementary Fig. S5a). Repeated t-SNE evaluations across varying perplexities and training seeds bound the projected silhouettes between 7.9×10^{-2} and 1.3×10^{-1} (Supplementary Fig. S5b). This pervasive low cluster separation persists across all representations, including raw model energies with distinct reference offsets, PCA-reduced SOAP summaries, and common 30-bin neighbor histograms (silhouette 5.6×10^{-2} before standardization).

These low silhouette metrics establish a specific physical mechanism. The divergent MD trajectories inhabit heavily overlapping local structural environments. Consequently, visual cluster separation reflects macroscopic trajectory divergence driven by cumulative force integration errors over time. This structural overlap mandates extracting fundamental model errors directly from explicit process observables. The projections identify macroscopic failure horizons, pointing to subtle force field inaccuracies that accumulate within a shared geometric phase space.

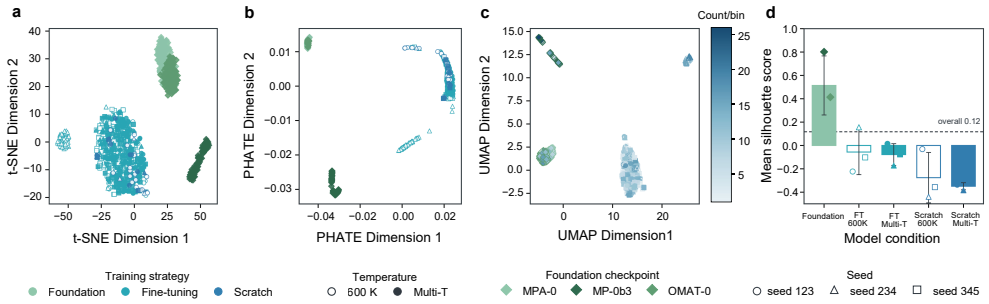

Fig. 6 Trajectory-feature projections across the 15-model MD comparison. a, t-SNE with perplexity 30 and random state 42. **b,** PHATE with random state 42. **c,** UMAP with 15 neighbors, minimum distance 0.1, and random state 42; colour intensity maps directly to local two-dimensional bin counts on a 45-bin histogram. **d,** Per-condition means of model-specific silhouette scores evaluated in the t-SNE plane, displaying individual checkpoint and seed symbols alongside a dashed overall mean of 1.2×10^{-1}. These metrics quantify clustering strictly within the lower-dimensional projection space. The input comprises 927 frames, ranging from 4 to 101 frames per model, and combines energies, Cartesian forces, neighbor histograms, and per-trajectory SOAP/PCA summaries. Model energy offsets, feature scaling, independent SOAP PCA bases, and unequal simulation failure horizons define the boundary conditions of the projected geometric layouts. Original-space separability and projection-sensitivity validations on the same dataset appear in Supplementary Fig. S5a,b and Supplementary Table S9, with constant columns excluded from all quantitative distance analyses.

2.6 Error surrogates expose the instability of learned feature associations

Because trajectory projections confirm that diverging models occupy shared structural environments, their kinetic failures must originate from how they internally map these geometries to predictive forces. Error surrogates evaluate this intrinsic mapping by ranking the learned feature associations. Feature rankings vary heavily across training seeds and model conditions (Fig. 7a,b). Te–Te and Cr-containing descriptors emerge as the leading features for fine-tuned checkpoints, whereas scratch conditions emphasize distinct combinations of host and dopant environments. The between-seed variation observed across identical architectures confirms that models capture structural behavior through degenerate, distributed feature representations. This indicate the attribution of macroscopic migration fidelity to a single dominant causal channel.

The cross-validation diagnostic yields R^2 values of $0.98 \pm 3 \times 10^{-3}$ for FT-600K, $0.97 \pm 3 \times 10^{-3}$ for FT-MultiT, $0.98 \pm 6 \times 10^{-3}$ for Scratch-600K, and $0.95 \pm 2.2 \times 10^{-2}$ for Scratch-MultiT (Supplementary Table S11). These metrics quantify the in-distribution fidelity of the five-fold diagnostic surrogate across the specific training seeds. Furthermore, the final higher-capacity TreeSHAP estimator operates on the full common sample to provide a closed-set mapping of the internal representations, establishing a localized interpretability baseline for the observed training distributions.

A supplementary coordinate-perturbation study evaluates the local relaxation response of native MACE inputs across six dopant–host combinations. This evaluation confirms the physical-input robustness of the models and establishes that macroscopic migration errors arise from complex, multi-channel interactions acting collectively across the structural descriptors.

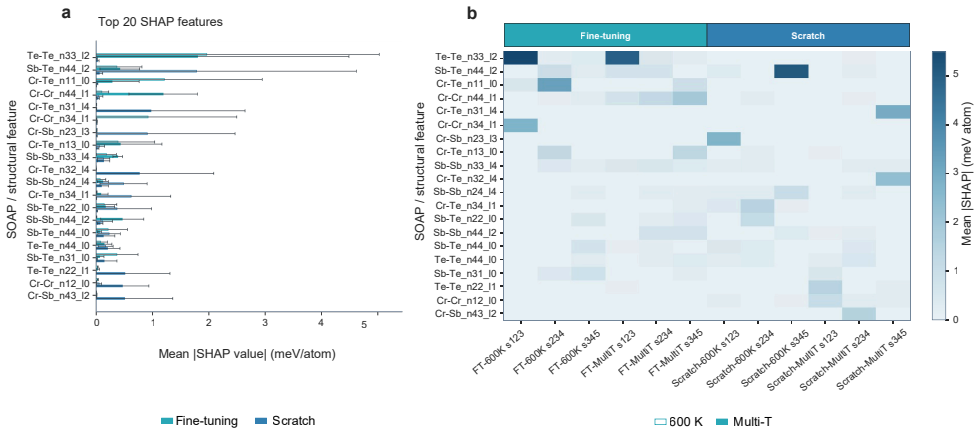

Fig. 7 Structural predictors of energy error depend on model condition and training seed. TreeSHAP explains gradient-boosted surrogates of the absolute energy error per atom on one common 267-frame test trajectory. **a,** Mean absolute SHAP values for the 20 features selected by the shared global ranking, with mean and sample standard deviation over training seeds 123, 234 and 345 within each of four conditions. Strategy colour distinguishes fine-tuning and scratch training; open/filled bars distinguish 600 K and MultiT. **b,** The same 20 feature rows aligned across all 12 model–seed columns on a shared meV·atom^{-1} scale. Species pairs and n_1n_2, l identify SOAP power-spectrum channels; they are descriptor indices, not isolated bonds or MACE neurons. Surrogates use 407 computed features and frame-aligned errors. Fidelity of the separate five-fold diagnostic estimator is given in Supplementary Table S11; it does not make the full-data TreeSHAP fit an out-of-sample causal explanation.

3 Discussion

When does a foundation model help? In this study, the answer depends on the physical observable being tested. The zero-shot checkpoints sustain the longest MD trajectories, whereas target adaptation minimizes held-out force errors and improves specific fixed-image energy alignments. Neither benefit guarantees the other. Within-QL distortions expose massive energy overestimations in foundation and fine-tuned models, while scratch models exhibiting high static accuracy collapse rapidly during MD. The central lesson is clear: a transferable prior and a trustworthy physical simulation are entirely different achievements. A foundation model delivers practical value only when explicitly validated for the intended dynamic process.

These separated failure modes dictate a clear order for model validation. Candidate models must first demonstrate static accuracy against first-principles references across the specific coordination changes relevant to the process. Validation must then proceed to self-driven structural relaxations and short MD trajectories to directly test stability and convergence. Finally, collective displacements and endpoint-return checks impose necessary periodic constraints that local thermal sampling leaves untested. Success at any single stage validates only that specific regime, confirming that orthogonal observables require independent verification.

Distribution shift offers a useful framework for these failures, but its application requires a precise physical definition. Adaptation datasets routinely miss process-relevant transition geometries even within matching chemical and temperature ranges.

Model-driven trajectories actively exacerbate this shift by exploring configurations absent from the reference set. Furthermore, discrepancies between pretraining and target electronic-structure protocols inject systemic calibration errors. Because foundation pretraining datasets are massive, researchers must isolate specific physical mismatches rather than relying on broad numerical distance metrics. Macroscopic model failure fundamentally arises from the interplay of these nested data shifts.

This targeted divergence provides a clear objective for active learning. Optimal data acquisition must focus on configurations where self-driven model rollouts expose unstable forces or unphysical topological profiles. Subsequent model updates demand rigorous evaluation against held-out dynamic tests alongside strict non-regression checks for previously stable trajectories. This feedback loop directly governs the choice between continued adaptation and specialist training. While feature attributions and ensemble disagreements can nominate candidate structures, their true acquisition value relies on proven performance gains under a fixed first-principles computational budget.

These findings establish a strict evaluation standard for MACE implementations in Cr-doped Sb_2Te_3, providing a clear methodological blueprint for broader dopant-host systems and alternative neural architectures.[R1-A4] The topological sensitivity of the diagnostic pathways focuses macroscopic evaluations strictly on structural mechanisms, rather than predicting absolute kinetic rates. Ultimately, a foundation model helps when its zero-shot robustness withstands the specific dynamic process under investigation, and target adaptation helps only when localized static improvements translate reliably into stable macroscopic simulations.

4 Conclusion

The deployment of MLIPs in materials discovery exposes a disconnect between static accuracy and dynamic reliability. By evaluating foundation, fine-tuned, and scratch models across distinct physical tasks in Cr-doped Sb_2Te_3, we demonstrate that these capabilities represent orthogonal dimensions of model performance. Models exhibiting high static fidelity on fixed-reference structures frequently collapse during unconstrained MD. Conversely, zero-shot foundation models that sustain prolonged dynamics severely mischaracterize localized transition barriers and collective periodic registries. Trajectory projections and feature attributions reveal the underlying physical mechanism for these divergences. Macroscopic model failures emerge from subtle force integration errors accumulating within shared geometric environments, driven by highly degenerate and unstable learned feature representations.

These findings establish a methodological mandate for the materials modeling community. The evaluation of universal priors and specialized potentials must advance beyond static benchmarking datasets. A reliable force field requires hierarchical validation that tests self-driven structural relaxations, transition pathway topologies, and macroscopic collective responses. Ultimately, AI-driven materials discovery demands process-specific dynamic validation. A transferable prior accelerates research only when its robustness withstands the precise physical process under investigation, ensuring that localized static accuracy translates reliably into faithful macroscopic simulations.

5 Methods

5.1 Reference data and model lineages

The reference calculations and AIMD data utilize PBE-based Quantum ESPRESSO for Cr-doped Sb_2Te_3 [68–70]. The 600 K target pool exactly matches the thermostat temperature of the MD simulations. The MultiT pool broadens thermal coverage to test temperature transferability to the target trajectories and process configurations. Path-specific computational settings and convergence states are detailed in Supplementary Sec. 1.

The MD comparison evaluates the MPA-0, MP-0b3, and OMAT-0 zero-shot foundation checkpoints alongside three distinct seeds for each of four trained conditions: Scratch-600K, FT-600K, Scratch-MultiT, and FT-MultiT. Fine-tuned grouped-A2 models initialize from an OMAT-FT checkpoint; scratch models deploy the registered $L_{\max} = 1$ architecture. Legacy pathway checkpoints appear separately in Fig. 4a–f and Fig. 5b–e. Figure 5a and Supplementary Fig. S1a,b summarize the evaluation and training workflows.

Training, validation, and test partitions are split by complete source trajectories to prevent adjacent-frame leakage across data splits.[R1-C10] Within each temperature regime, scratch and fine-tuned models share identical target-data splits and use seeds 123, 234, and 345. The 600 K pair uses 1,786/311/267 training/validation/test configurations, while the MultiT pair uses 2,659 training and 767 test configurations. This design enables a direct evaluation of training routes at a fixed 600 K baseline and under broadened thermal coverage. An overlap audit of 25 legacy migration images confirms the absence of exact or threshold-defined SOAP near-duplicates within the audited training files.[R1-A2] The separate 89-configuration Scratch-5% control serves as a data-volume sensitivity baseline.[R1-A5] The overall route comparison evaluates the integrated macroscopic effects of pretraining, architecture, and optimization. Finally, the force-supervised training protocol enforces a physical non-zero force requirement, admitting exactly zero forces exclusively for isolated-atom energy references.[R1-C8]

5.2 Process observables and uncertainty

Fixed-path scoring evaluates every model on identical reference geometries, whereas self-relaxed scoring initializes an NEB path and evolves intermediate geometries driven by the model's own forces. These operators interrogate distinct physical properties, as self-relaxation simultaneously alters the sampled configurations and the resulting energy landscape. For a reference image set $\Gamma_{\text{ref}} = \{X_i\}$, the fixed-path forward peak is:

$$B_M^{\text{fixed}}(\Gamma_{\text{ref}}) = \max_i \{E_M(X_i) - E_M(X_0)\}. \tag{1}$$

The diagnostic discrepancy $|B_M^{\text{fixed}} - B_{\text{ref}}|$ quantifies activation-energy errors strictly when the electronic reference, pathway optimization, and endpoint definitions are fully converged. Self-relaxed evaluations compute $\Gamma_M = \text{NEB}_M(X_0, X_1)$ instead. Validating a self-relaxed physical barrier requires explicit structural correspondence alongside the peak energy. The fixed and self-relaxed operators independently measure supplied-path

calibration and model-driven structural optimization; their respective peak energies characterize fundamentally distinct physical responses.[R1-A1]

Each MD run initiates from the identical 2,050-atom state at a simulation clock of 200 ps and targets a total simulated time of 1 ns.[R1-C9] The timestep is 1 fs, trajectory output occurs every 200 steps, the thermostat target is 600 K, the Langevin friction parameter is 0.01, and the MD seed is 1100. Trajectory durations are reconstructed directly from recorded steps, with stopped runs terminating precisely at their physical failure points. Sample standard deviations across training seeds quantify checkpoint initialization variance, while foundation model variation reflects differences across distinct pretrained architectures.

5.3 Diagnostic analysis and reproducibility

The trajectory-feature analysis embeds 927 sampled frames from 15 MD model conditions using t-SNE, PHATE, and UMAP. Quantitative distances map macroscopic output features, with any constant-valued placeholders rigorously excluded prior to spatial evaluation.[R1-C12] Original-space scores are calculated before and after standardization, and t-SNE projections are repeated at perplexities 5, 30, and 50 with nominal seeds 0, 42, and 123. A common neighbour-histogram analysis ensures a shared physical basis across the divergent models, providing an objective structural comparison framework alongside the independently fitted SOAP PCA coordinates.

The SHAP analysis evaluates a separate, common 267-frame, 86-atom held-out trajectory across all four trained conditions and their respective seeds. The target metric is the absolute energy error per atom; inputs comprise 390 periodic SOAP features and 17 structural descriptors. A gradient-boosted regressor with 200 estimators, a learning rate of 0.05, a depth of 8, a subsampling rate of 0.8, and random state 42 supplies the full-sample TreeSHAP values. A separate 50-estimator, depth-4 surrogate utilizes shuffled five-fold validation to rigorously quantify in-distribution diagnostic fidelity across the sampled trajectory. Full definitions and per-seed metrics are detailed in Supplementary Sec. 6.

Data availability

The figure-associated numerical tables, model and dataset identities, convergence status records, and diagnostic outputs accompany the submission source package. The project repository is available at https://github.com/yicao-elina/MigrationBench. Figure-specific source identities are explicitly detailed in the Supplementary Data to ensure exact checkpoint reproducibility.

Code availability

The accompanying source package provides the executed scripts and frame-aligned inputs for the SHAP analysis and feature audit, alongside the scripts and compact source tables required to independently reproduce all figure evidence.[R1-C13]

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Author contributions

Y.C.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, and writing (original draft, review, and editing). P.M.: conceptualization, literature investigation, and writing (review and editing). Y.W.: visualization and writing (review and editing). V.W. and J.P.: investigation and data curation (DFT NEB computations). P.C.: conceptualization, resources, supervision, project administration, funding acquisition, and writing (review and editing).

Competing interests

The authors declare no competing interests.

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Supplementary Information: When Does a Foundation Model Help? Process-Specific Reliability of Machine-Learned Interatomic Potentials

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1 Reference data, training routes and pointwise evaluation

1.1 Reference scope and computational provenance

The reference archive compiles Quantum ESPRESSO calculations utilizing PBE-based electronic structure for Cr-doped Sb_2Te_3 [1, 2]. The parent AIMD methods define a baseline protocol comprising D3 dispersion, SG15 pseudopotentials, $1.36 \times 10^3/5.44 \times 10^3$ eV wavefunction/charge-density cutoffs, cold smearing of 0.20 eV, an electronic threshold of 1.36×10^{-5} eV, and thermal sampling at 300, 600, and 1200 K. To ensure exact computational provenance, configuration-specific input parameters take precedence over these general parent settings for all migration and sliding profiles. Consequently, intermediate pathways extracted from earlier profiles are evaluated as explicit structural sequences, remaining geometrically distinct from fully optimized DFT minimum-energy paths.

The evaluation framework separates the dataset into three structurally distinct reference collections. The legacy five-image paths (G1, G2, and Q1–Q3) exclusively support the initial grouped overlap and retraining controls. The canonical 81-atom, ten-image pathways supply the primary migration evaluation, establishing precise in-gap, gap-to-QL, and within-QL reference peaks of 1.52, 0, and 1.84 eV, respectively.

Finally, the fixed sliding reference profiles establish peaks of 1.65 eV for the pristine system and 2.57 eV for interstitial Cr in the gap, accompanied by separate endpoint-return checks. Because these image sets map distinct geometric sequences, the legacy G1 benchmark (0.34 eV) and the canonical in-gap reference function as independent diagnostic targets and are not interchangeable.

1.2 Model matrix and training-data comparisons

The contemporary pointwise evaluation examines Scratch-600K, FT-600K, Scratch-MultiT, and FT-MultiT configurations, each utilizing independent training seeds 123, 234, and 345. The 600 K dataset matches the thermostat temperature governing the MD evaluations. The MultiT dataset integrates configurations across multiple temperatures to assess how broadened thermal coverage influences target-temperature dynamics and process stability. Within each regime, scratch and fine-tuned architectures operate on identical target-data splits: the 600 K pair uses a 1,786/311/267 configuration split for training/validation/test, whereas the MultiT pair utilizes 2,659 training and 767 test configurations.

This cross-regime evaluation assesses the integrated macroscopic impact of dataset scale and thermal diversity on model performance. Accordingly, the plotted energy records—comprising pooled arrays of 5,358/801 and 7,977/2,301 entries—represent the aggregate of three seed predictions per unique configuration, effectively quantifying initialization variance.

Fine-tuned models derive from the grouped-A2, OMAT-FT-start, stage-two compiled lineage. Scratch models in the primary evaluation deploy the registered $L_{\max} = 1$ architecture. The legacy pathway campaigns evaluate distinct checkpoint architectures to capture operational variations. Cryptographic SHA256 hashes for all model and data files are cataloged in the accompanying source registry to guarantee exact reproducibility. This comparative design evaluates the complete, integrated performance of these practical training routes, encompassing the combined influences of pretraining, neural architecture, and optimization parameters.

Table S1 Current uncentred held-out errors. Energy RMSE in meV atom^{-1} and Cartesian force-component RMSE in meV Å^{-1}, reported as means and sample standard deviations over training seeds 123, 234, and 345. The underlying 24 train/test evaluations are retained in the current-release source table. Because the thermal conditions utilize distinct test compositions, their respective metrics independently characterize model fidelity within each specific data domain.

Model	Test frames / seed	Energy	Force
Scratch-600K	267	30 ± 3	344 ± 36
FT-600K	267	41 ± 5	260 ± 6
Scratch-MultiT	767	30 ± 3	414 ± 192
FT-MultiT	767	33 ± 1	233 ± 2

The parity visualization subtracts a DFT trajectory-cluster mean from both the reference and predicted energies. Shared train/test clusters utilize one pooled reference mean, whereas clusters exclusive to the test split use their specific test reference

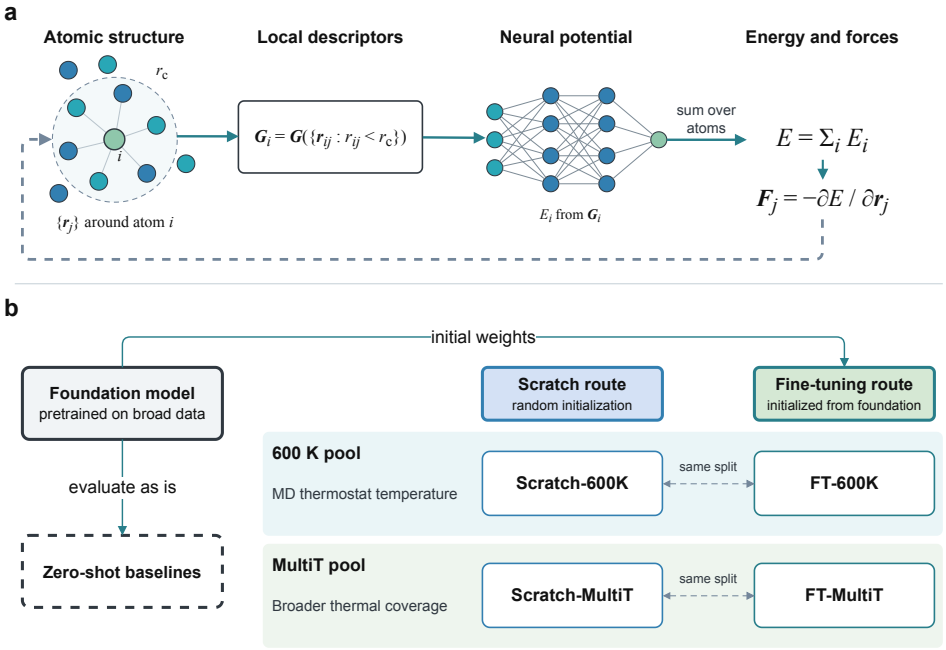


Fig. S1 Training routes and comparison design. a, Mapping from atomic structure through local descriptors and a neural potential to the total energy, with forces given by its negative coordinate gradient. **b,** The zero-shot foundation baselines and the four trained conditions, arranged by training route (scratch or fine-tuning) and target-data regime (600 K or MultiT). The 600 K pool matches the MD thermostat temperature; MultiT broadens thermal coverage. Within each regime, the two routes use the same target-data split. Because the thermal regimes utilize distinct dataset scales and test compositions, and the model implementations natively differ in architecture and optimization, this comparative design evaluates the integrated macroscopic performance of the complete training routes. Model identities accompany the numerical results.

mean. This cluster-centering transformation normalizes the visual display to highlight relative correlations; deployed models natively evaluate uncentered thermodynamic states. While absolute residuals remain invariant under this uniform shift, the resulting cluster-centered correlations specifically capture local energy fluctuations within each trajectory space. Consequently, the quantitative errors in Table S1 are calculated directly from the raw uncentered predictions. For visualization clarity, the Scratch-MultiT force parity plot truncates 106 outlier components beyond ± 10 eV Å^{-1}, whereas the quantitative error calculations retain the complete unclipped dataset.

1.3 Force-label audit

To guarantee the integrity of force-supervised training, we implement a strict data validation protocol. A valid force-bearing training export requires explicit, verified force arrays; energy-only geometric exports are programmatically prevented from registering as force-labelled frames.

The comprehensive audit covers 9,163 records across historical FT-600K, historical FT-MultiT, grouped FT-600K, and grouped Scratch-5% partitions. Every admitted record contains a validated non-zero force field, with the explicit exception of nine single-atom Cr, Sb, or Te `IsolatedAtom` energy-reference configurations which intrinsically possess zero forces. The historical training cohorts (1,891 and 3,015 configurations for FT-600K and FT-MultiT, respectively) remain distinct from the current grouped datasets. The manifest explicitly records frame identities, source headers, force-array presence, source-file checksums, and split membership. Model provenance is established strictly through these explicit cryptographic checksums rather than generic family labels.

1.4 Grouped splitting, overlap and data-volume controls

The grouped partitioning protocol isolates complete source trajectories entirely within training, validation, or test sets. This design explicitly prevents trajectory-adjacent frames from leaking across partitions, ensuring robust out-of-sample evaluation.

The structural audit evaluates 25 legacy migration images against the full 600 K grouped training file and the 89-configuration Scratch-5% subset. This evaluation confirms zero exact global matches and zero exact Cr-local matches. Furthermore, no SOAP cosine distance falls at or below the 10^{-4} threshold. Minimum observed SOAP distances are 5.4×10^{-4} and 6.4×10^{-4}, respectively. The minimum comparable Cr-local RMSD is 2.06 Å, which exceeds the 0.05 Å near-duplicate threshold. While species-compatible RMSD comparisons identify a limited candidate pool, the SOAP evaluation comprehensively covers all audited images, confirming the structural independence of the test partitions.

The legacy G1 grouped fixed-image discrepancies stabilize at 0.47 ± 0.12 eV for FT-600K, 1.20 ± 0.27 eV for full Scratch, and 0.68 ± 0.20 eV for Scratch-5%. This statistically controlled multi-seed benchmark replaces earlier preliminary point estimates. The full-data scratch and fine-tuned implementations (1,786 configurations) supply the strictly volume-controlled primary comparison, while the 89-frame subset is retained exclusively to evaluate data-volume sensitivity under extreme scarcity.

Figure S2 summarizes the legacy G1 discrepancies and the threshold-normalized overlap audit.

Table S2 Legacy grouped control, kept distinct from canonical pathways. Values represent the mean ± sample standard deviation across three independent training seeds (except for the single MPA0 checkpoint). G1 is a 61-atom, five-image reference with a QE forward peak of 0.34 eV. The final column reports diagnostic fixed-image discrepancies rather than dynamic physical barriers. The current pointwise release in Table S1 supplies freshly recomputed static metrics, which replace legacy rounded values to maintain a consistent independent G1 control.

Route	Train frames	G1 discrepancy (eV)	Training seeds
FT-600K	1,786	0.47 ± 0.12	3
Scratch	1,786	1.20 ± 0.27	3
Scratch-5%	89	0.68 ± 0.20	3
MPA0	zero-shot	0.24	checkpoint

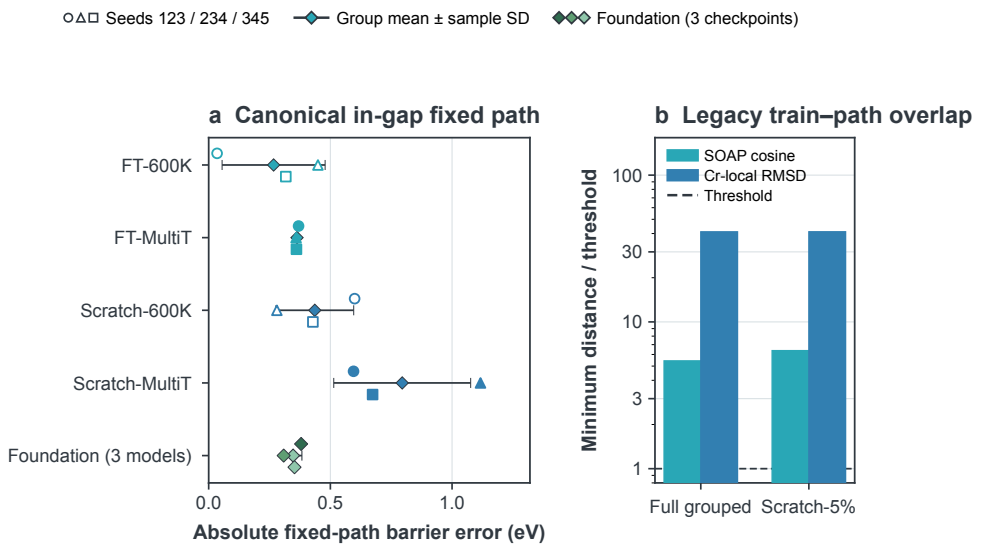


Fig. S2 Legacy grouped retraining and train–path overlap audit. a, Absolute fixed-image discrepancies on G1 for FT-600K, Scratch, and Scratch-5%. The plot reports individual training seed outcomes alongside their mean ± sample standard deviation. The foundation symbol represents the single zero-shot checkpoint. The dashed 0.16 eV baseline marks the superseded preliminary point estimate. **b,** Minimum training-set SOAP and composition-compatible local RMSD distances, normalized by their respective near-duplicate thresholds. Values strictly greater than one confirm the structural separation of the test images from the training distribution. The G1 and 25-image audits function explicitly within the legacy reference collection, remaining methodologically distinct from the 1.52 eV canonical in-gap benchmark evaluated in the main text.

2 Migration references, operators and seed-resolved discrepancies

2.1 Metric definitions

For image set $\Gamma = \{X_0, \ldots, X_n\}$, let

$$\Delta E_M(i) = E_M(X_i) - E_M(X_0), \tag{1}$$

$$B_M^{\rightarrow}(\Gamma) = \max_i \Delta E_M(i), \tag{2}$$

$$B_M^{\leftarrow}(\Gamma) = \max_i E_M(X_i) - E_M(X_n). \tag{3}$$

Including the initial image ensures a nonnegative forward peak. Consequently, a monotonically downhill profile yields $B^{\rightarrow} = 0$, independent of its endpoint energy or reverse peak. Because a minimal peak discrepancy can coincide with an inaccurate overall profile, whole-band spans, forward peaks, and reverse peaks are evaluated as structurally distinct quantities.

Fixed scoring evaluates identical geometries across all calculators. Endpoint-initialized, self-relaxed NEB conversely evaluates a model-optimized image set Γ_M.

Optimizer convergence confirms the internal mathematical minimization of the target model; establishing physical pathway correspondence requires independent structural verification against the DFT reference. Accordingly, the main figures report explicit structural diagnostic values, maintaining a strict separation from kinetic rate-constant derivations.

2.2 Current main-figure matrix and validation protocol

The main seed matrix comprises 15 model rows and eight tasks, yielding 120 finite numerical records. An independent validation protocol computes all absolute discrepancies against the task references, establishing full mathematical parity within the evaluated matrix. The three migration references are explicitly defined as 1.52, 0, and 1.84 eV, while the sliding references are 1.65 and 2.57 eV. These precise numerical definitions anchor the figure-associated tables.

To guarantee rigorous data integrity, the matrix records are cryptographically joined against the raw Scratch-600K campaign, matching all self-relaxed records by seed, pathway, and model SHA256 hashes. Topological states are explicitly evaluated based on their fully relaxed geometries. For instance, the structural optimization of the seed 123/within-QL configuration establishes a precise finite forward peak of 2.35 eV. Main Fig. 5c,d reflects these exact topological states, scaling the self-relaxed axis to retain the complete data distribution without omitting any finite values.

The finalized diagnostic registry documents all 120 validated pathway entries. Because optimizer convergence signifies local force minimization, all raw source files, hash-matched logs, and explicit structural classifications are retained to ensure complete methodological transparency.

Finally, the dataset maintains explicit lineage tracking. Fig. 4a–f visualizes a representative profile ensemble: the scratch trajectory serves as a legacy baseline, the foundation band aggregates multiple baseline checkpoints, and the fine-tuned curves sample distinct architectural members. To ensure exact reproducibility, specific checkpoint provenance is tracked exclusively via cryptographic identities rather than generic family labels.

Table S3 Checkpoint ensemble composition for selected profile visualizations. Counts indicate the number of distinct model checkpoints evaluated for each pathway class in main Fig. 4a–f. Foundation counts represent distinct zero-shot architectures, whereas scratch and fine-tuned counts represent independent training initializations. For the FT-MultiT condition, the within-QL self-relaxed pathway yields zero stable trajectories under the strict topological criteria. The visualized ensemble explicitly focuses on these core conditions, maintaining geometric consistency separate from the expanded contemporary seed matrix.

Operator / path	Foundation	Legacy Scratch	FT-600K	FT-MultiT
Fixed / in gap	9	3	3	2
Fixed / gap to QL	9	3	3	2
Fixed / within QL	9	3	3	2
Self / in gap	9	3	3	3
Self / gap to QL	9	3	3	3
Self / within QL	9	3	3	0

Table S4 Canonical fixed-path reference discrepancies. Absolute energy discrepancies relative to the canonical task peaks (eV). Trained conditions report the mean ± sample standard deviation across three independent training seeds; foundation entries represent individual zero-shot checkpoints. Statistics are derived from the comprehensive 120-record matrix detailed in Fig. 5b. Migration benchmarks are evaluated directly on the fully relaxed first-principles coordinates. Sliding discrepancies serve as localized scalar metrics, complementing the comprehensive full-profile and endpoint-return structural diagnostics.

Model	In gap	Within QL	Pristine sliding	Cr sliding
MP-0B3	0.38	22.56	0.60	0.64
OMAT-0	0.31	22.20	0.28	0.66
MPA0	0.35	24.70	0.41	0.59
FT-600K	0.27 ± 0.21	10.91 ± 1.16	0.50 ± 0.34	0.41 ± 0.25
FT-MultiT	0.36 ± 0.01	11.00 ± 0.46	0.71 ± 0.43	0.60 ± 0.50
Scratch-600K	0.44 ± 0.16	0.45 ± 0.05	0.63 ± 0.31	1.48 ± 1.05
Scratch-MultiT	0.80 ± 0.28	0.75 ± 0.39	0.24 ± 0.20	0.98 ± 0.49

Table S5 Legacy QE pathway references. Forward and reverse peak values are evaluated explicitly relative to their respective endpoints. These first-principles pathway calculations establish rigorous physical baselines, maintaining clear methodological separation from the subsequent model-relaxed trajectories. This legacy collection remains structurally distinct from the canonical main-figure reference set.

Name	Class	N_{img}	$\mathrm{QE}^{\rightarrow}_{\mathrm{ref}}$ (eV)	$\mathrm{QE}^{\leftarrow}_{\mathrm{ref}}$ (eV)
G1	within-gap	5	0.34	0.02
G2	within-gap	5	0.58	0.02
Q1	gap-to-QL/into-QL	5	2.79	1.02
Q2	gap-to-QL/into-QL	5	4.00	3.95
Q3	gap-to-QL/into-QL	5	1.10	0.75
M1	mixed	5	1.03	0.21
L1	within-QL	5	1.76	0.99
C1	within-QL	10	1.03	1.03
C2	within-QL	10	1.56	2.11
C3	within-QL	10	1.69	1.67
C4	within-QL	10	2.06	1.54
C5	within-QL	10	1.41	1.07

The Q1 Scratch discrepancy demonstrates stability across the three independent training seeds (0.51 ± 0.11 eV), whereas Q2 and Q3 exhibit higher initialization variance. Conversely, the corresponding FT-600K discrepancies consistently stabilize near 10 eV across equivalent evaluations. This multi-seed resolution confirms that these specific error signatures represent reproducible characteristics of the respective training routes rather than statistical artifacts. These macroscopic evaluations specifically quantify fixed-image coordinate sensitivity, operating independently of the dynamic kinetic models.

Table S6 Legacy migration pathway registry. Class- and geometry-based nomenclatures directly map the archived pathway controls to their published release identifiers. This explicit taxonomy maintains clear geometric separation between the five-image G/Q baseline references and the ten-image canonical pathways evaluated in the primary text.

Name	Class	Atoms	Path ID	Classification basis
G1	within-gap	61	`1-4`	Cr stays in the vdW gap between sites defined as in-gap in the path queue; image-set reference path
G2	within-gap	61	`1-7`	Cr stays in the vdW gap between sites defined as in-gap in the path queue
Q1	gap-to-QL/into-QL	61	`1-2`	Path queue sites place Cr moving from the vdW gap into the quintuple layer
Q2	gap-to-QL/into-QL	61	`1-3`	Path queue sites place Cr moving from the vdW gap into the quintuple layer
Q3	gap-to-QL/into-QL	61	`1-5`	Path queue sites place Cr moving from the vdW gap into the quintuple layer; the queue asks for a geometry recheck
M1	mixed	61	`1-6`	Exact image coordinates give a mixed gap/QL transition
L1	within-QL	61	`1-8`	Exact image coordinates place Cr within the quintuple layer
C1	within-QL	81	`neb_1`	Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain
C2	within-QL	81	`neb_2`	Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain
C3	within-QL	81	`neb_3`	Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain
C4	within-QL	81	`neb_4`	Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain
C5	within-QL	81	`neb_5`	Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain

2.3 Additional dopant responses

The supplementary Cu, Li, and Na profiles extend the primary Cr-doped Sb_2Te_3 evaluation framework to assess structural responses across alternative dopant chemistries. These first-principles pathways establish explicit physical baselines for diverse transition-metal and alkali substitutions. Because the primary investigation precisely isolates data-volume, initialization, and architectural effects exclusively within the Cr system, these supplementary profiles serve to contextualize chemical response mechanisms. They

Table S7 Path-resolved fixed-path errors for the grouped models. Absolute errors (eV), mean ± sample standard deviation across three independent training initializations for FT-600K, Scratch-5%, and full Scratch. The QE reference column gives the forward energy reference of each image set. Q1–Q3 are UNCONVERGED paths; their entries are image-set energy spans used for fixed-path comparison, not physical migration barriers. Errors are fixed-path errors on identical images, reported per path and not pooled across classes. Path names are defined in Table S6.

Path	Class	QE reference	FT-600K	Scratch-5%	Scratch
G1	within-gap (reference path)	0.34	0.47 ± 0.12	0.68 ± 0.20	1.20 ± 0.27
G2	within-gap	0.58	0.21 ± 0.18	0.25 ± 0.21	0.33 ± 0.06
Q1	gap-to-QL/into-QL	2.79	9.60 ± 0.57	0.33 ± 0.31	0.51 ± 0.11
Q2	gap-to-QL/into-QL	4.00	10.40 ± 0.44	1.22 ± 0.66	0.72 ± 0.55
Q3	gap-to-QL/into-QL	1.10	9.70 ± 0.20	1.07 ± 0.31	1.77 ± 1.00

establish local geometric fidelity across distinct chemical environments while explicitly bounding the macroscopic model-selection framework to the primary Cr-doped host.

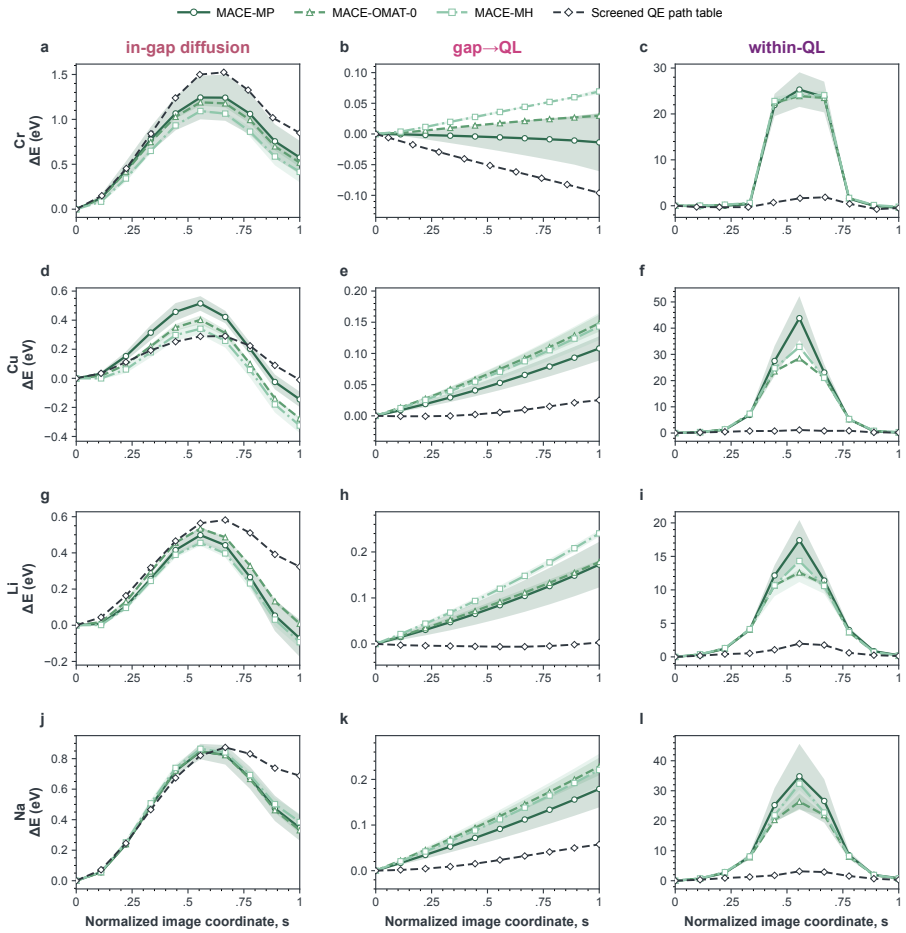


Fig. S3 Canonical foundation-model pathway responses for four dopants in Sb_2Te_3. Rows detail migration pathways for Cr (**a**–**c**), Cu (**d**–**f**), Li (**g**–**i**), and Na (**j**–**l**); columns map in-gap, gap-to-QL, and within-QL configurations. Curves evaluate the MACE-MP, MACE-OMAT-0, and MACE-MH checkpoint groups against the established first-principles baselines (dashed lines). Energies are referenced to the initial image, with the horizontal coordinate representing the normalized reaction coordinate. Shaded bands quantify the aggregate variance across available model checkpoints. These first-principles benchmarks establish exact physical energy landscapes, evaluating base-model structural fidelity across diverse local chemical domains.

3 Collective sliding

Collective sliding profiles quantify macroscopic mechanical responses by evaluating prescribed interlayer displacements for pristine and Cr-intercalated structures. Fixed-geometry evaluations benchmark static thermodynamic assignments against the canonical first-principles coordinates, whereas endpoint-initialized, model-relaxed bands evaluate dynamic structural optimization. For all profiles, the first-image energy is independently centered to establish a uniform baseline.

To guarantee absolute thermodynamic consistency, endpoint-return fidelity is enforced and evaluated alongside transition-peak accuracy. Rigorous structural verification—encompassing periodic boundary conditions, explicit species mapping, and global energy references—ensures that any observed endpoint divergences accurately reflect intrinsic model parameterizations (such as spatial interaction range limitations) rather than compositional artifacts. Figure S4 summarizes these validated fixed and model-relaxed mechanical profiles.

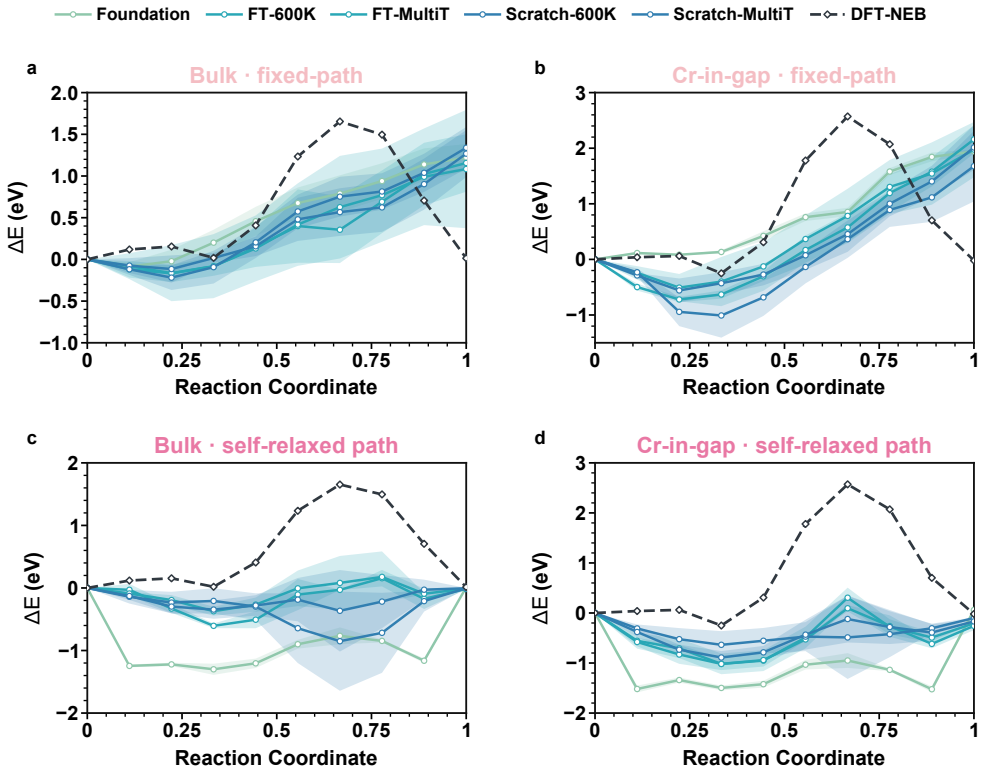


Fig. S4 Macroscopic sliding profiles across foundation, fine-tuned, and scratch routes. a,b, Fixed-path relative energies for pristine and Cr-in-gap Sb_2Te_3, respectively. **c,d,** Endpoint-initialized, model-relaxed dynamic profiles for the corresponding systems. Solid coloured distributions distinguish the evaluated training conditions and their intra-route initialization variance; the dashed dark curve represents the canonical first-principles reference. The normalized coordinate maps the progression of the structural phase space. First-principles reference peaks establish absolute macroscopic sliding barriers of 1.65 eV (pristine) and 2.57 eV (Cr-in-gap). This evaluation framework rigorously decouples static peak calibration, global profile curvature, and thermodynamic endpoint-return fidelity, directly benchmarking the intrinsic dynamic optimization capabilities of the respective neural potentials.

4 Matched MD protocol and trajectory horizons

To ensure rigorous dynamic consistency, all simulations initiate from a shared macroscopic state saved at step 200,000, featuring finite momenta in a 2,050-atom periodic cell. The global simulation target is a 1 ns clock (step 1,000,000). Consequently, a trajectory terminating at 236 ps represents exactly 36 ps of dynamic continuation beyond the common 200 ps initialization. The protocol specifies a 1 fs timestep, a 600 K thermostat target, a Langevin friction parameter of 0.01, and trajectory sampling every 200 steps. The MD seed (1100) uniquely defines the thermostat initialization and operates independently of the model training seeds. Trajectories are recorded precisely to their observed physical stability limits (horizons); the foundation checkpoints successfully maintain stability through the full 1 ns target.

Table S8 Explicit MD horizons across evaluated models. Clock times precisely track the simulated duration from the shared 200 ps initialization. Foundation entries represent distinct zero-shot checkpoints, whereas trained-family entries evaluate the dynamic variance across three independent training seeds. Evaluated horizons exactly reflect the unextrapolated limits of structural stability.

Checkpoint / training seed	Clock (ps)	Continuation (ps)	Outcome
Foundation_MPA0	1000	800	Target reached
Foundation_MP0b3	1000	800	Target reached
Foundation_OMAT0	1000	800	Target reached
FT600K_A2_seed123	236	36	Threshold reached
FT600K_A2_seed234	1000	800	Target reached
FT600K_A2_seed345	485	285	Threshold reached
FTMultiT_A2_seed123	237.2	37.2	Threshold reached
FTMultiT_A2_seed234	270.2	70.2	Threshold reached
FTMultiT_A2_seed345	286	86	Threshold reached
Scratch600K_Lmax1_seed123	204.4	4.4	Threshold reached
Scratch600K_Lmax1_seed234	200.8	0.8	Threshold reached
Scratch600K_Lmax1_seed345	200.4	0.4	Threshold reached
ScratchMultiT_Lmax1_seed123	201.2	1.2	Threshold reached
ScratchMultiT_Lmax1_seed234	200.4	0.4	Threshold reached
ScratchMultiT_Lmax1_seed345	200.4	0.4	Threshold reached

Thermodynamic observables are extracted directly from the native dynamics. Kinetic temperature is derived from explicitly recorded momenta, while hydrostatic pressure is evaluated as $-\mathrm{tr}(\sigma)/3$ via the saved stress tensor. In-plane mean squared displacement (MSD) incorporates fractional-coordinate unwrapping within the triclinic cell, and the velocity autocorrelation function utilizes multiple time origins normalized by the zero-lag baseline.

Pair coordination numbers are integrated from the partial radial distribution functions (RDF) up to the invariant first-minimum cutoffs of 3.2 Å (Cr–Sb), 4.1 Å (Cr–Te), and 4.2 Å (Sb–Te). The global all-atom RDF processes ordered pair counts using $r_{\max} = 10$ Å, 0.1 Å resolution, and a 20-frame sampling stride. Visualizations utilize localized smoothing for clarity, while all quantitative integrations employ the raw, unmodified pair-count arrays. Because dynamic stability horizons intrinsically vary across the evaluated models, these structural summaries precisely integrate over the explicit stable observation window of each respective trajectory.

Because the fine-tuned and scratch trajectories are intrinsically failure-bounded, macroscopic transport derivations (e.g., long-time diffusion coefficients or Green–Kubo thermal conductivities) are explicitly excluded to maintain strict physical validity.

5 Current trajectory-feature analysis

5.1 Feature definition and physical boundaries

The structural and thermodynamic phase space is mapped using 927 sampled frames across the 15 evaluated models (ranging from 4 to 101 frames per architecture). The state of each frame is defined by a 6,191-dimensional explicit output vector, which concatenates the total energy, 6,150 Cartesian force components, a 30-bin neighbor-distance histogram, and ten localized SOAP PCA components.

To rigorously bound the interpretability framework, this analysis explicitly operates on these macroscopic output features rather than internal neural hidden states. Because absolute model energies retain architecturally distinct reference offsets, and the SOAP PCA bases are trajectory-specific, these concatenated vectors serve to map relative topological clusters rather than a universal physical coordinate system. Conversely, the shared 30-bin neighbor histogram provides an absolute, structurally invariant basis for cross-trajectory Euclidean comparisons.

Table S9 Original-space topological metrics for the 927-frame ensemble. Silhouette labels correspond to the 15 model identities visualized in the main text. Standardization applies column-wise z-scoring across the aggregate diagnostic sample to normalize explicit feature scales. The invariant neighbor-distance bins isolate structural separation independently of the trajectory-specific PCA bases. Scores quantify topological clustering within the explicitly sampled population horizons.

Feature space	Dimensions	Silhouette
Original output features (6191)	6191	6.3×10^{-2}
Original output features, cosine	6191	6.1×10^{-2}
Z-scored output features	6191	-5.9×10^{-2}
Common neighbour histogram (30)	30	5.6×10^{-2}
Z-scored neighbour histogram	30	5.2×10^{-2}

5.2 Projection sensitivity

The displayed t-SNE utilizes perplexity 30 and random state 42. PHATE employs random state 42 with its default neighbor setting; UMAP utilizes 15 neighbors, a minimum distance of 0.1, Euclidean distance, and random/transform seeds of 42. Panel c visualizes a 45-bin two-dimensional histogram count with a maximum capacity of 26, serving as a structural density map rather than a formal free-energy surface.

To quantify projection robustness, t-SNE is evaluated directly on the 6,191-column matrix across perplexities 5, 30, and 50, employing initialization seeds 0, 42, and

123 alongside PCA initialization. These algorithmic parameters evaluate visualization variance, yielding a silhouette range of 7.9×10^{-2}–1.3×10^{-1}. The observed sensitivity, coupled with the scale dependence of the original-space scores, confirms that visual clusters represent algorithm-dependent structural similarities rather than disjoint physical mechanisms. Figure S5 summarizes the scale and projection evaluations; the accompanying tables detail the exact quantitative metrics.

Table S10 Algorithmic *t*-SNE sensitivity metrics. Scores utilize the 15 model labels. Trustworthiness is evaluated using five neighbors, comparing each projection directly against the unprojected input matrix. These metrics explicitly quantify the topological stability of the dimensionality reduction.

Perplexity	Seed	Silhouette	Trustworthiness
5	0	7.9×10^{-2}	0.97
5	42	7.9×10^{-2}	0.97
5	123	7.9×10^{-2}	0.97
30	0	1.2×10^{-1}	0.98
30	42	1.2×10^{-1}	0.98
30	123	1.2×10^{-1}	0.98
50	0	1.3×10^{-1}	0.97
50	42	1.3×10^{-1}	0.97
50	123	1.3×10^{-1}	0.97

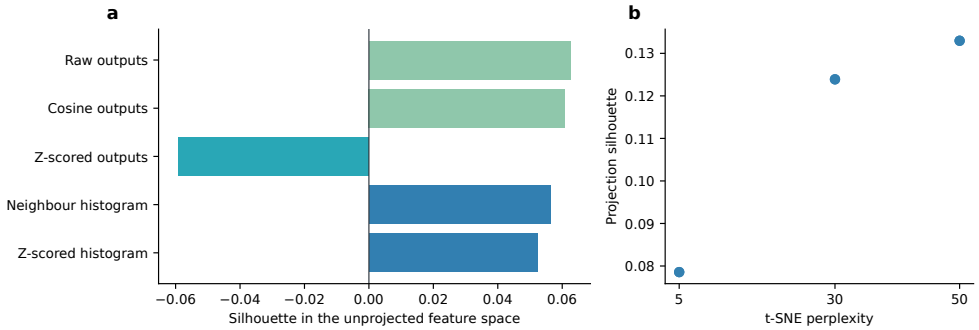


Fig. S5 Scale and projection metrics for the explicit feature dataset. a, Silhouette scores before projection for raw output features, standardized output features, and the invariant neighbor histogram. **b,** *t*-SNE silhouette values across distinct perplexities and algorithm initializations. Original-space values and per-run settings are detailed in Tables S9 and S10. The raw output-space score is 6.3×10^{-2} (standardized: -5.9×10^{-2}), while the common histogram score is 5.6×10^{-2}. These quantitative bounds establish the projection as a descriptive population diagnostic.

6 Common-configuration error surrogates and SHAP

The SHAP analysis evaluates surrogate error predictions on a shared 267-frame, 86-atom test trajectory (`2L_o3_t3_600K_aimd_1`), strictly matched via explicit frame alignment and exact reference-energy verification. All four model families and their respective three training seeds are evaluated on this identical configuration set. This cross-family analysis exclusively isolates the shared 600 K trajectory to maintain exact structural alignment across all evaluated architectures.

The target variable is the absolute total-energy error normalized by atom count (meV atom^{-1}). The input basis comprises 390 periodic, inner-averaged DScribe SOAP features for Cr/Sb/Te ($r_{\mathrm{cut}} = 5$ Å, $n_{\mathrm{max}} = 4$, $l_{\mathrm{max}} = 4$), concatenated with 17 explicit macroscopic structural features. SOAP power-spectrum labels identify specific chemical channels and radial/angular indices, providing an explicit mathematical basis for mapping local geometric sensitivity.

For the final TreeSHAP evaluation, each model–seed instance utilizes a gradient-boosting regressor (200 estimators, learning rate 0.05, maximum depth 8, subsampling fraction 0.8, random state 42). A localized five-fold KFold cross-validation (50 estimators, depth 4) quantifies diagnostic fidelity. These cross-validation metrics establish the in-distribution fidelity of the error surrogate on the sampled trajectory.

Table S11 Error-surrogate fidelity diagnostic. Values represent the mean ± sample standard deviation across the three underlying training seeds. Each entry summarizes the distinct five-fold diagnostic estimator on the identical 267-frame sample. The MAE is normalized to meV atom^{-1}. These metrics explicitly quantify in-distribution surrogate fit fidelity.

Model	Five-fold R^2	MAE (meV atom^{-1})
FT-600K	$0.98 \pm 3 \times 10^{-3}$	1.2 ± 0.1
FT-MultiT	$0.97 \pm 3 \times 10^{-3}$	1.3 ± 0.1
Scratch-600K	$0.98 \pm 6 \times 10^{-3}$	1.2 ± 0.1
Scratch-MultiT	$0.95 \pm 2.2 \times 10^{-2}$	1.7 ± 0.8

The primary visualization shares a global top-20 feature ordering across both panels. Bar plots aggregate the mean absolute SHAP value and sample standard deviation across the three training seeds within each condition, while the heat map details all 12 individual model–seed columns. The substantial between-seed variation observed across identical architectures confirms that models capture structural behavior through degenerate, distributed feature representations. This explicitly precludes the attribution of macroscopic migration fidelity to a single dominant causal channel.

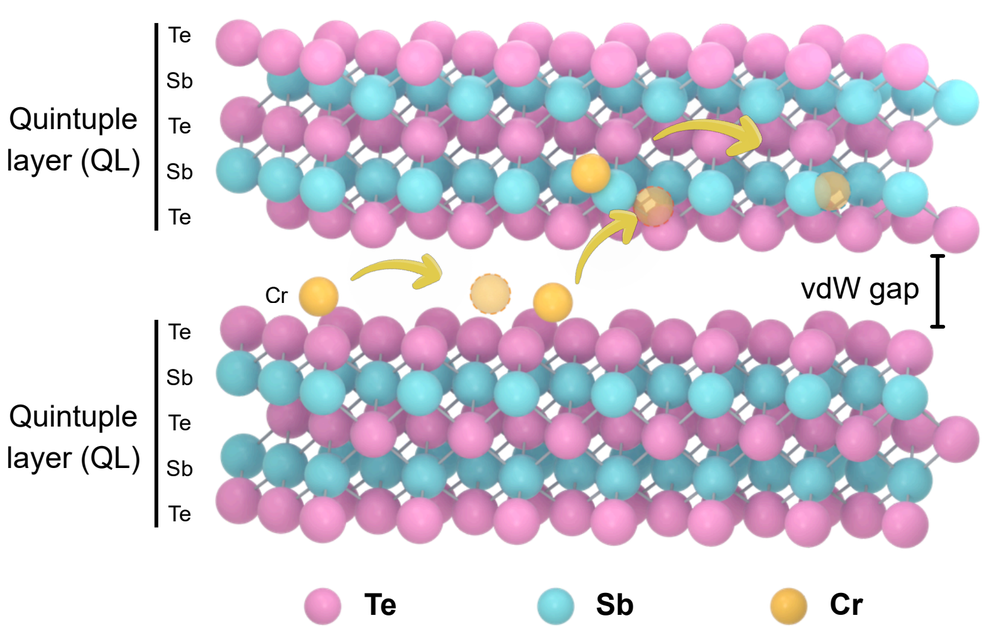
To ensure complete transparency, the accompanying executable workflow provides the fully reproducible pipeline. This module ingests the common XYZ geometries and evaluation tables, builds the 407 computed features, verifies structural alignment, fits the surrogates, and exports all per-seed metrics and attribution tables. The feature audit and figure-specific source ledger are included in the repository release.

References

[1] Giannozzi, P., Baroni, S., Bonini, N., Calandra, M., Car, R., Cavazzoni, C., Ceresoli, D., Chiarotti, G.L., Cococcioni, M., Dabo, I., *et al.*: Quantum espresso: a modular and open-source software project for quantumsimulations of materials. Journal of physics: Condensed matter **21**(39), 395502 (2009)

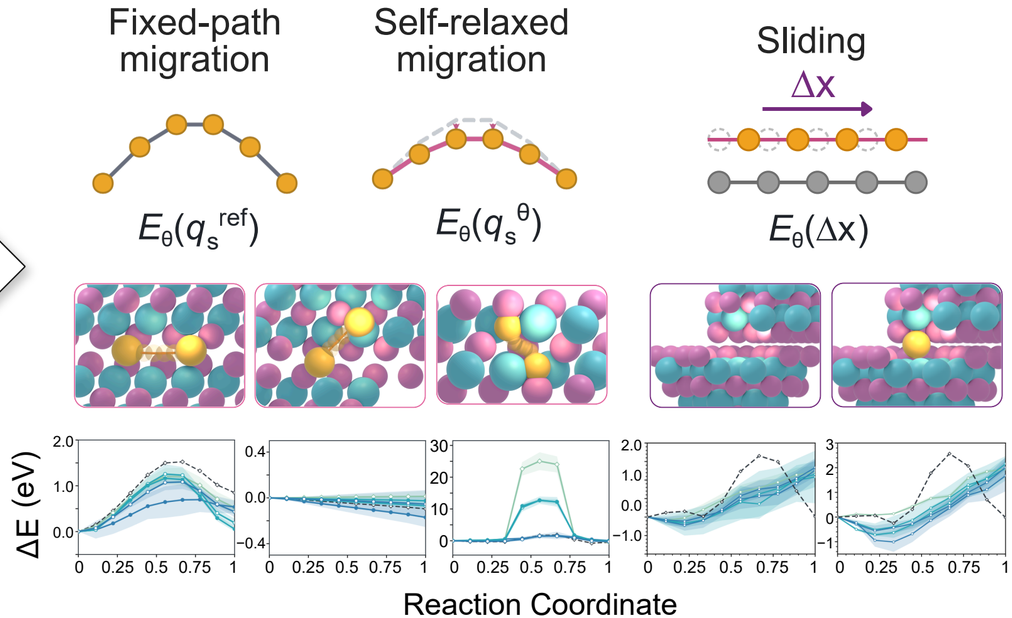
[2] Perdew, J.P., Burke, K., Ernzerhof, M.: Generalized gradient approximation made simple. Physical review letters **77**(18), 3865 (1996)

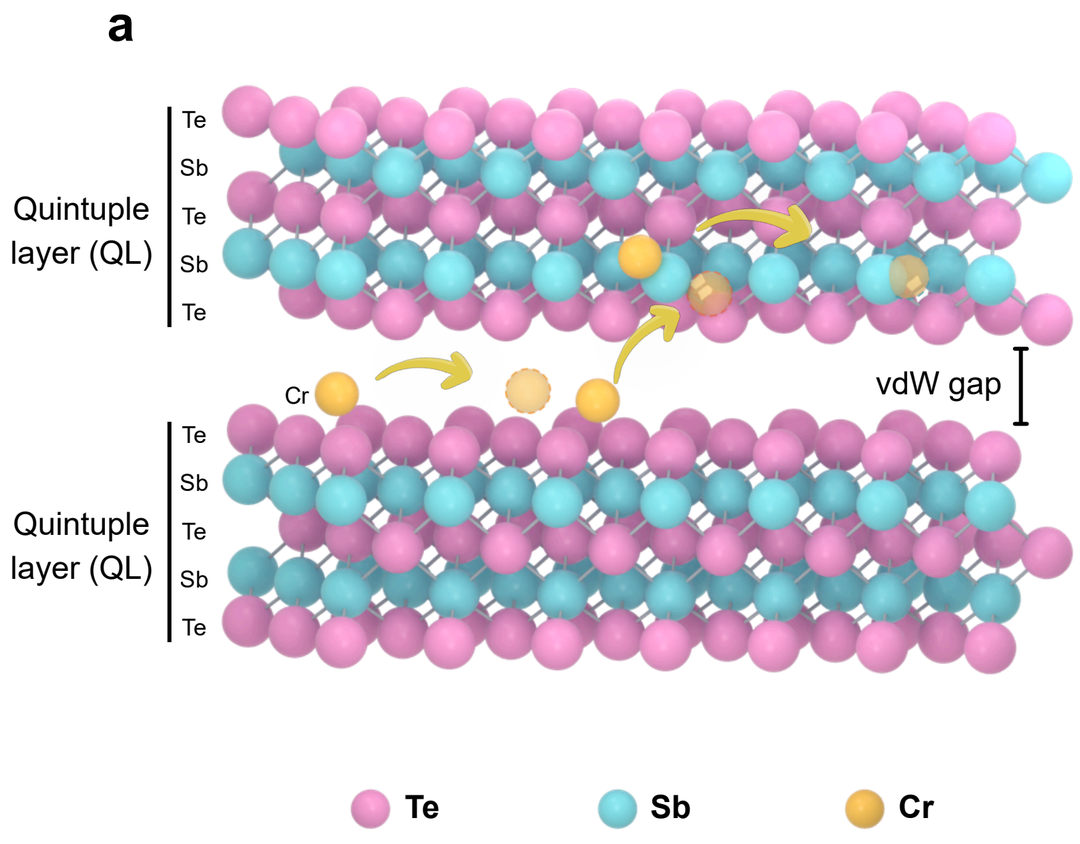
Do MLIP foundation models help with atomic migration simulation?

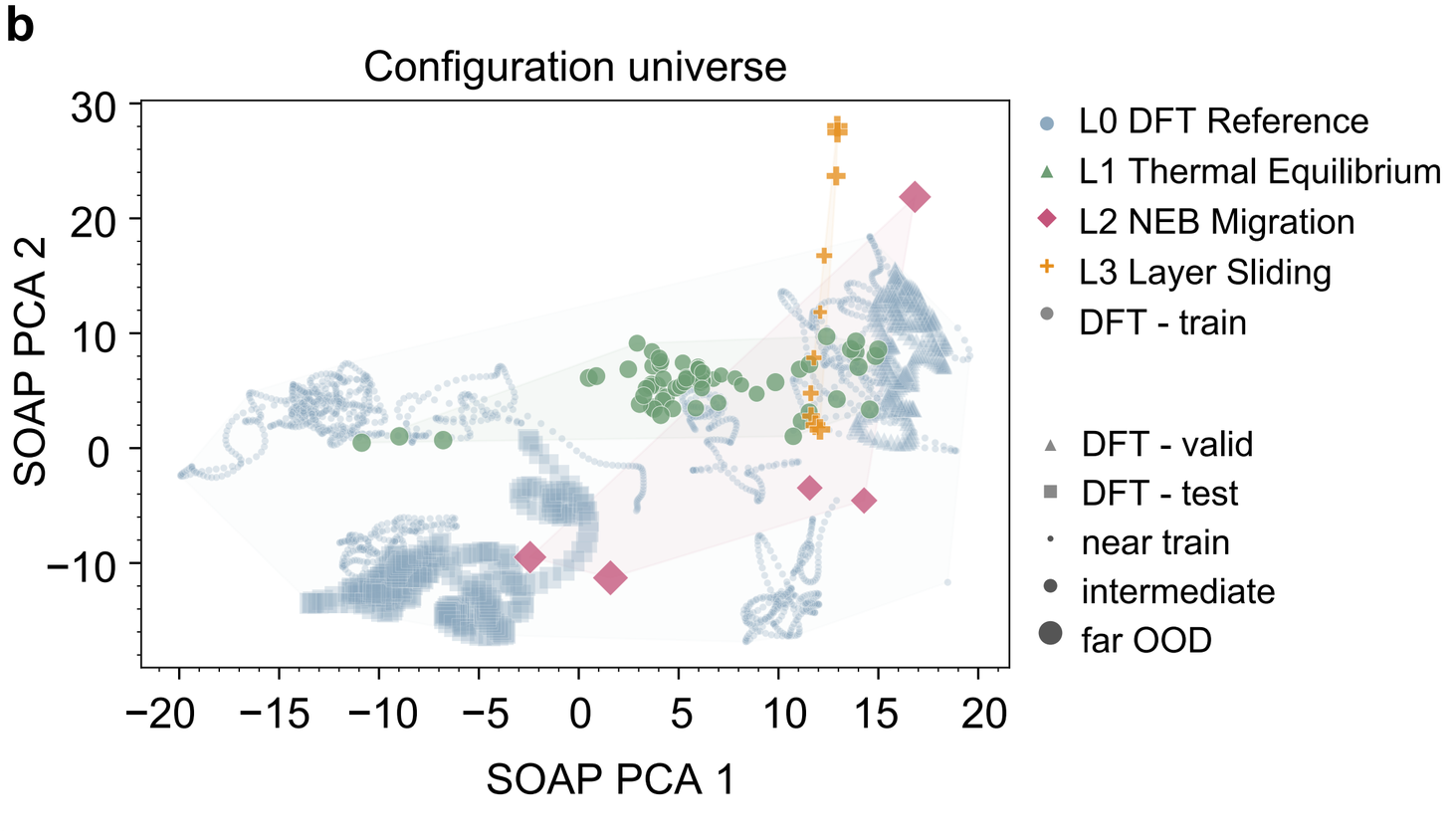


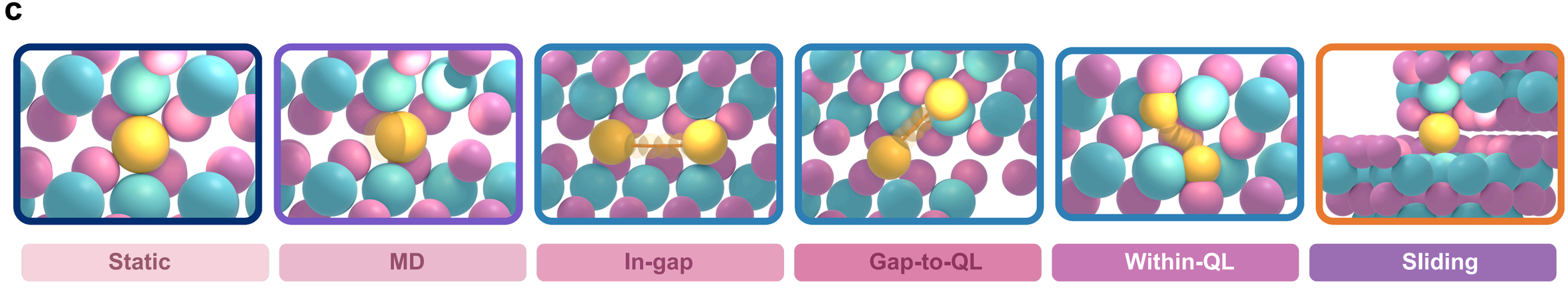
Energy prediction

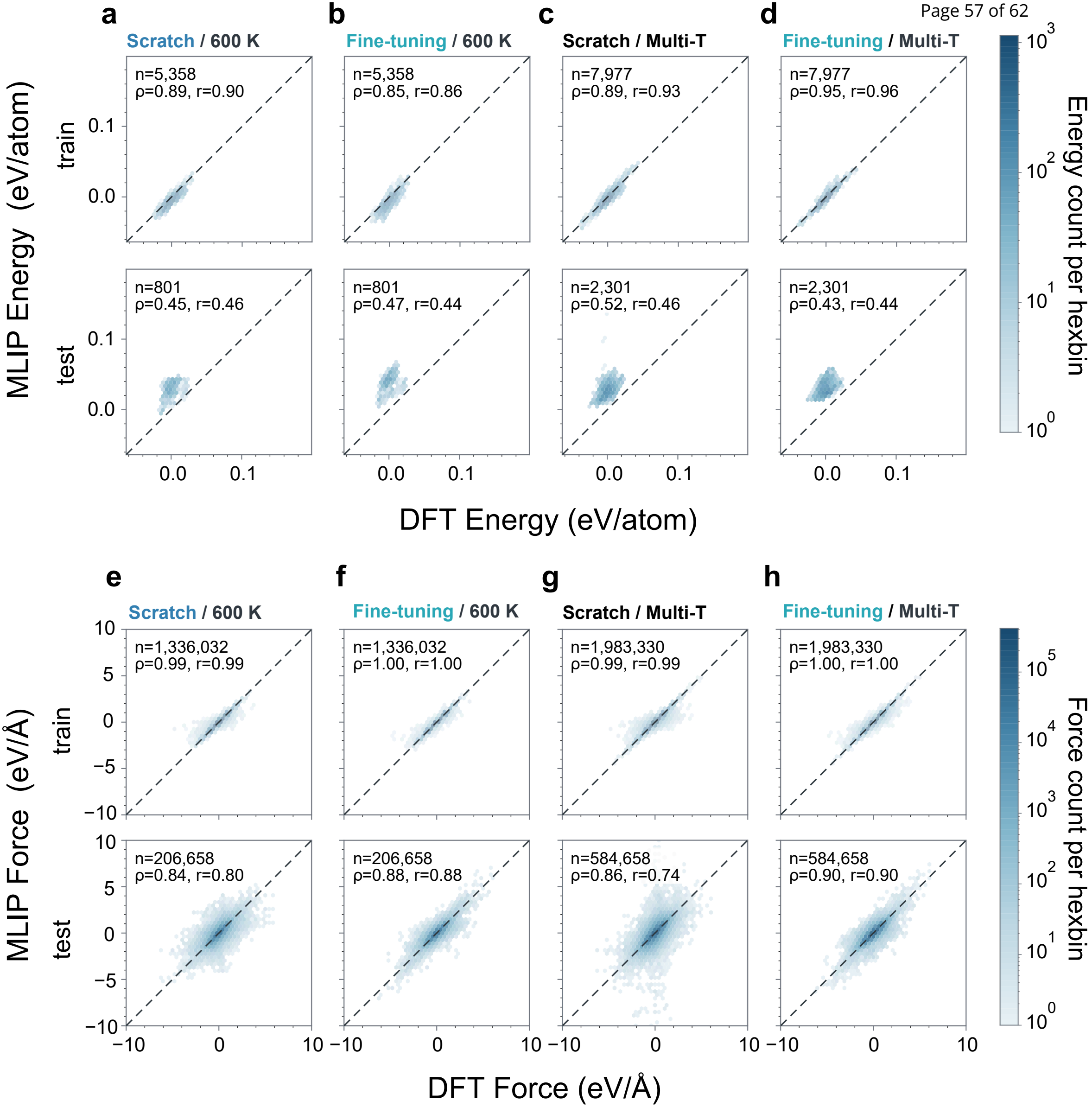
It depends on the specific process you care about.

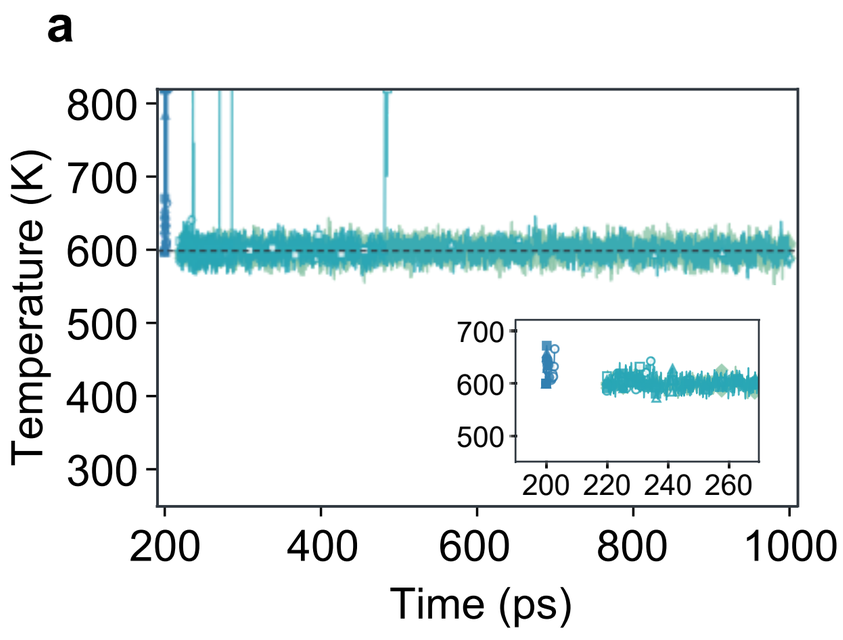


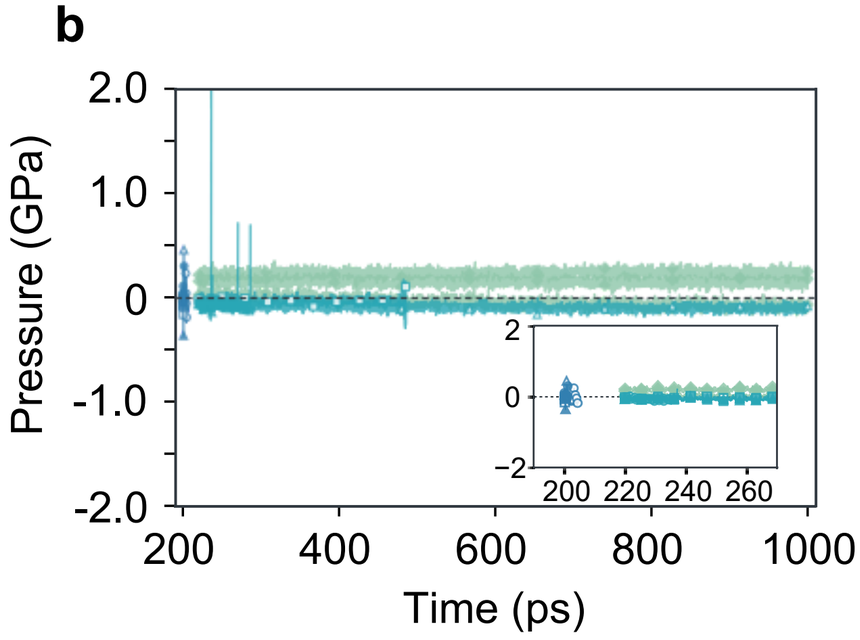

a
Quintuple layer (QL)
Te
Sb
Te
Sb
Te
Cr
vdW gap
Quintuple layer (QL)
Te
Sb
Te
Sb
Te
Te
Sb
Cr

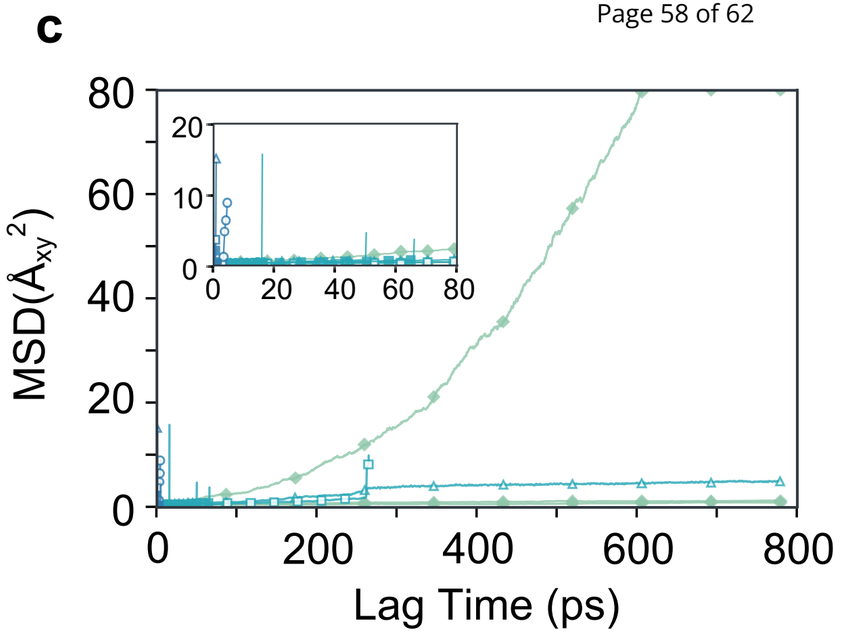

b
Configuration universe
SOAP PCA 2
SOAP PCA 1
30
20
10
0
−10
−20
−15
−10
−5
0
5
10
15
20
L0 DFT Reference
L1 Thermal Equilibrium
L2 NEB Migration
L3 Layer Sliding
DFT - train
DFT - valid
DFT - test
near train
intermediate
far OOD

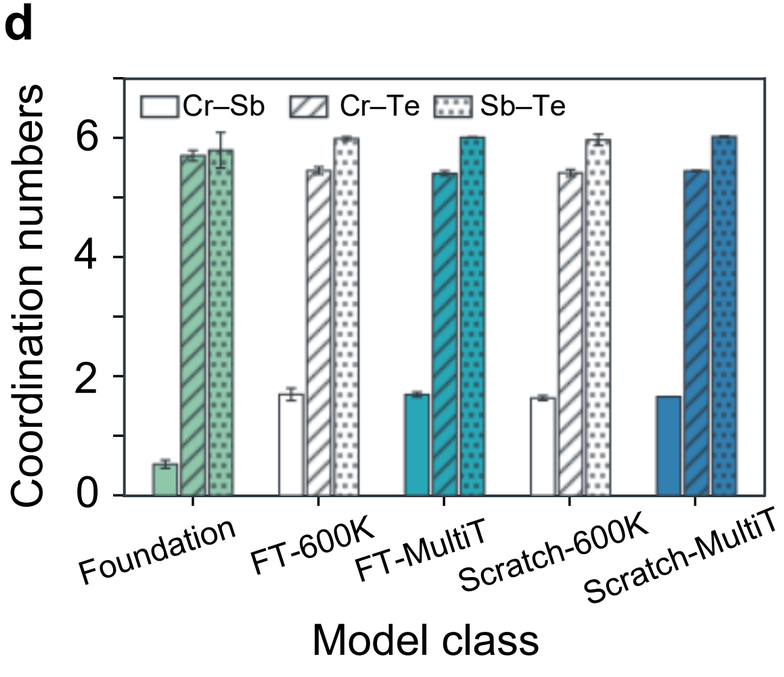

c
Static
MD
In-gap
Gap-to-QL
Within-QL
Sliding

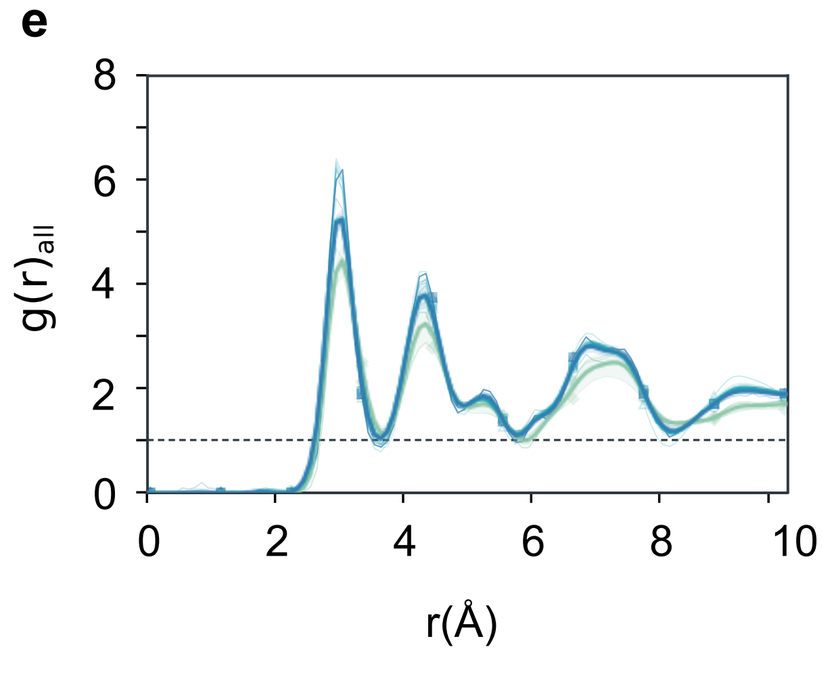

a
Scratch / 600 K
n=5,358
ρ=0.89, r=0.90
n=801
ρ=0.45, r=0.46
b
Fine-tuning / 600 K
n=5,358
ρ=0.85, r=0.86
n=801
ρ=0.47, r=0.44
c
Scratch / Multi-T
n=7,977
ρ=0.89, r=0.93
n=2,301
ρ=0.52, r=0.46
d
Fine-tuning / Multi-T
n=7,977
ρ=0.95, r=0.96
n=2,301
ρ=0.43, r=0.44
train
test
MLIP Energy (eV/atom)
DFT Energy (eV/atom)
Energy count per hexbin
e
Scratch / 600 K
n=1,336,032
ρ=0.99, r=0.99
n=206,658
ρ=0.84, r=0.80
f
Fine-tuning / 600 K
n=1,336,032
ρ=1.00, r=1.00
n=206,658
ρ=0.88, r=0.88
g
Scratch / Multi-T
n=1,983,330
ρ=0.99, r=0.99
n=584,658
ρ=0.86, r=0.74
h
Fine-tuning / Multi-T
n=1,983,330
ρ=1.00, r=1.00
n=584,658
ρ=0.90, r=0.90
MLIP Force (eV/Å)
DFT Force (eV/Å)
Force count per hexbin

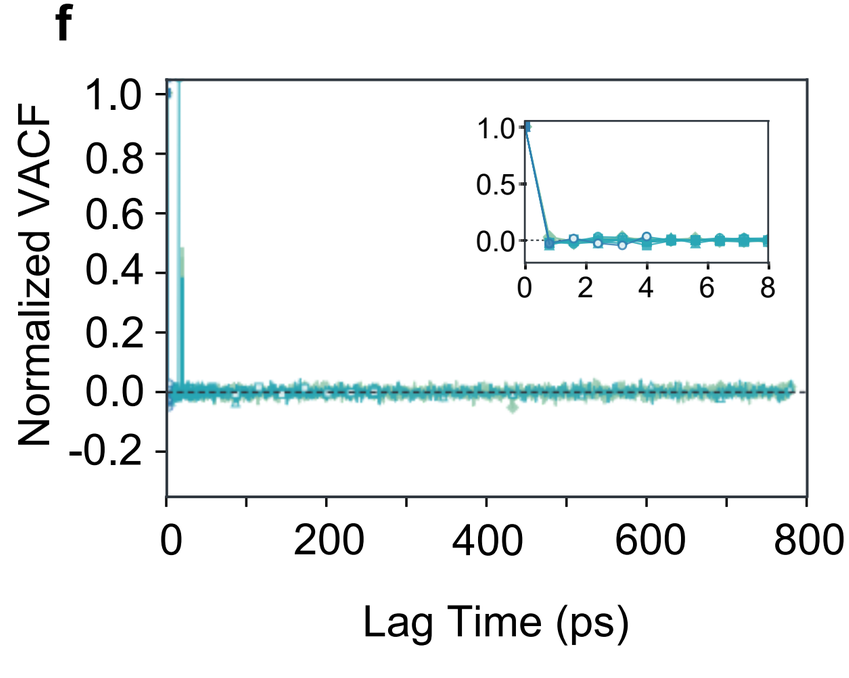




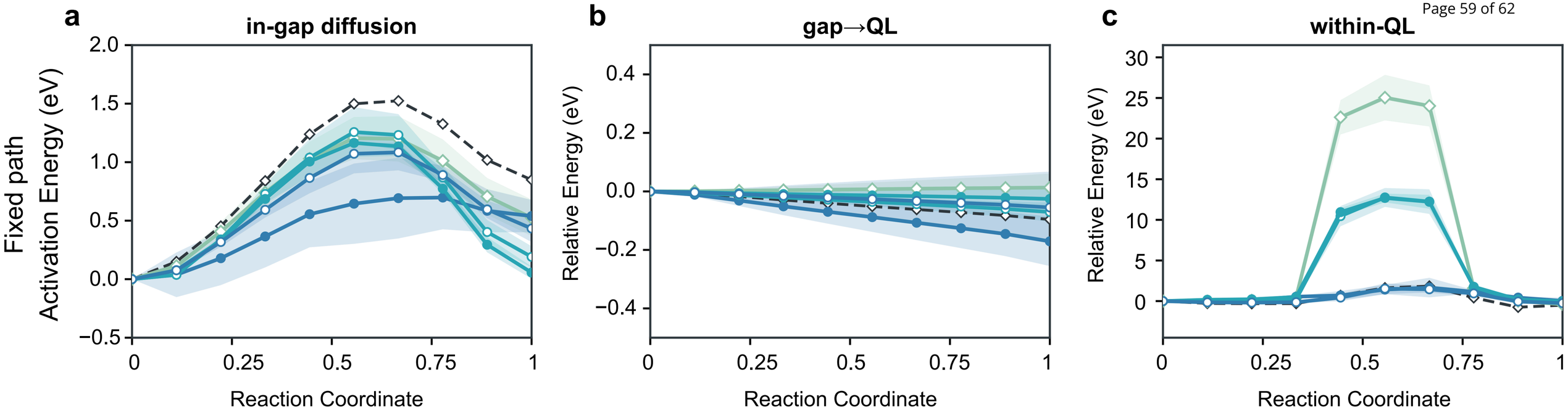


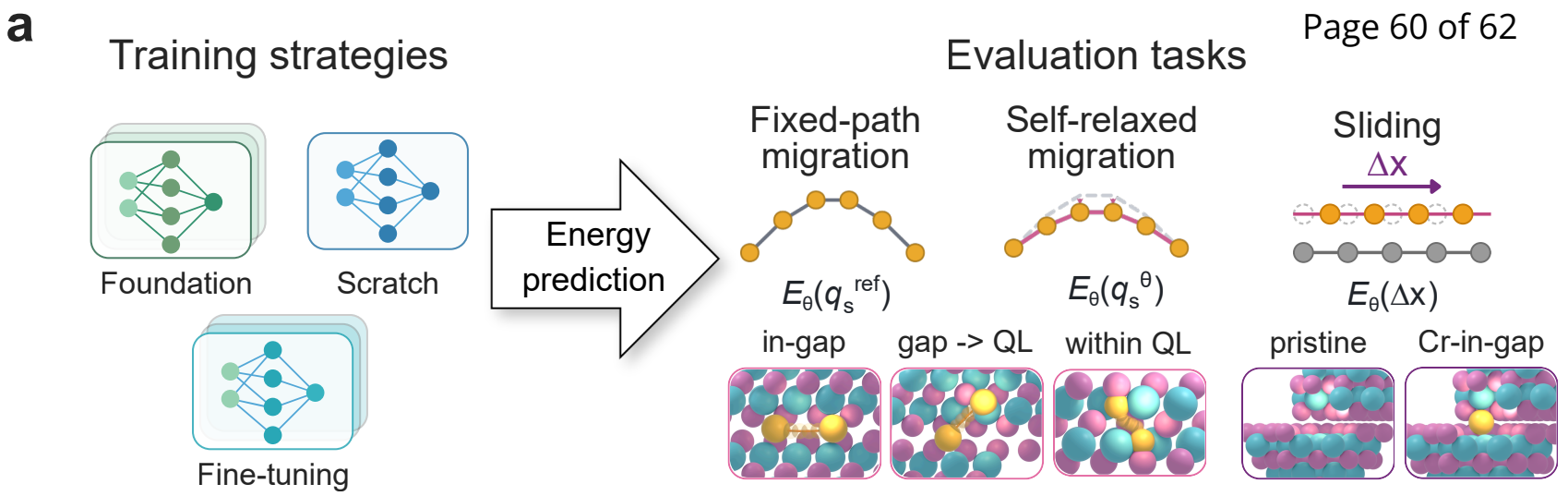


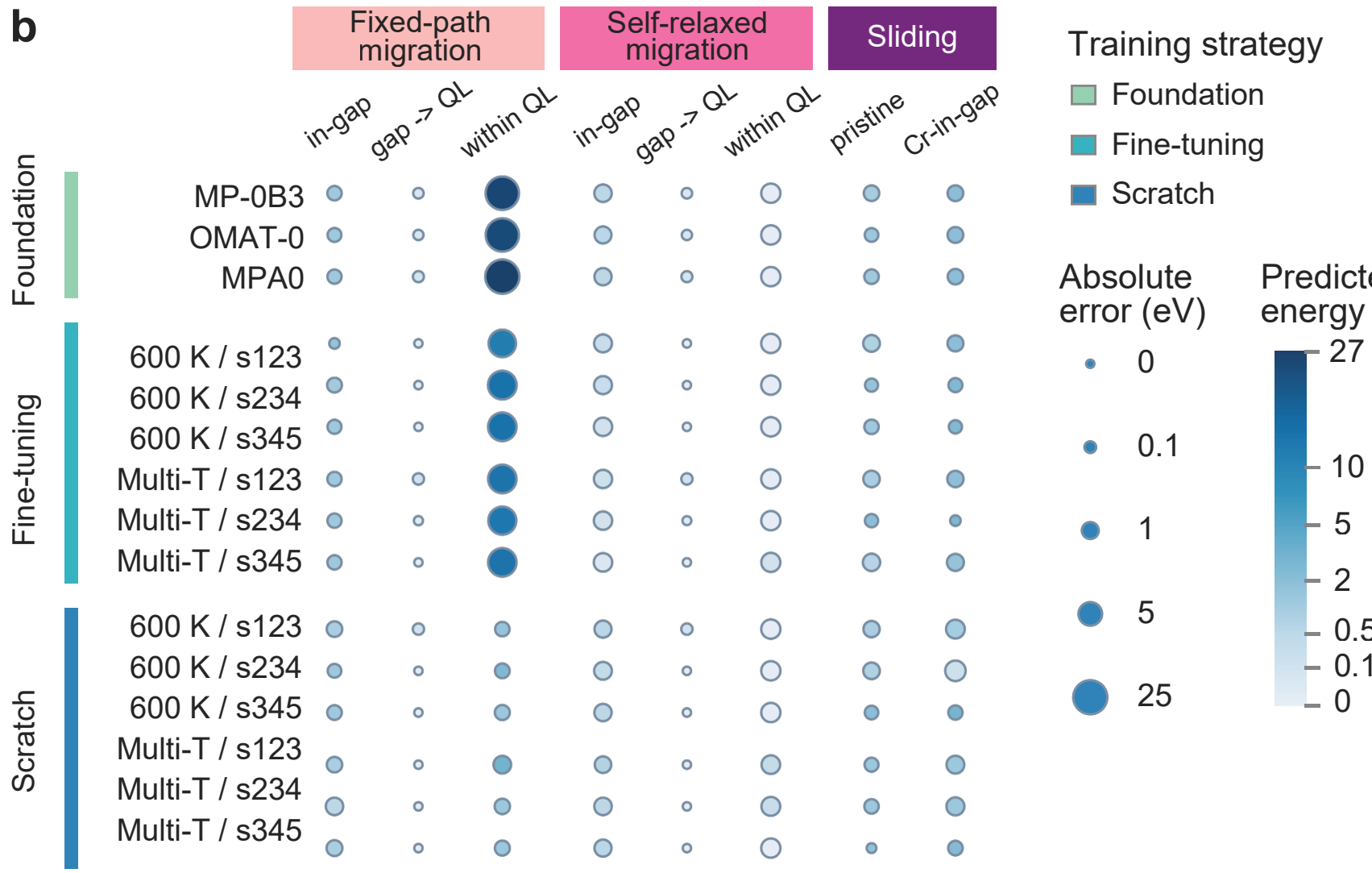


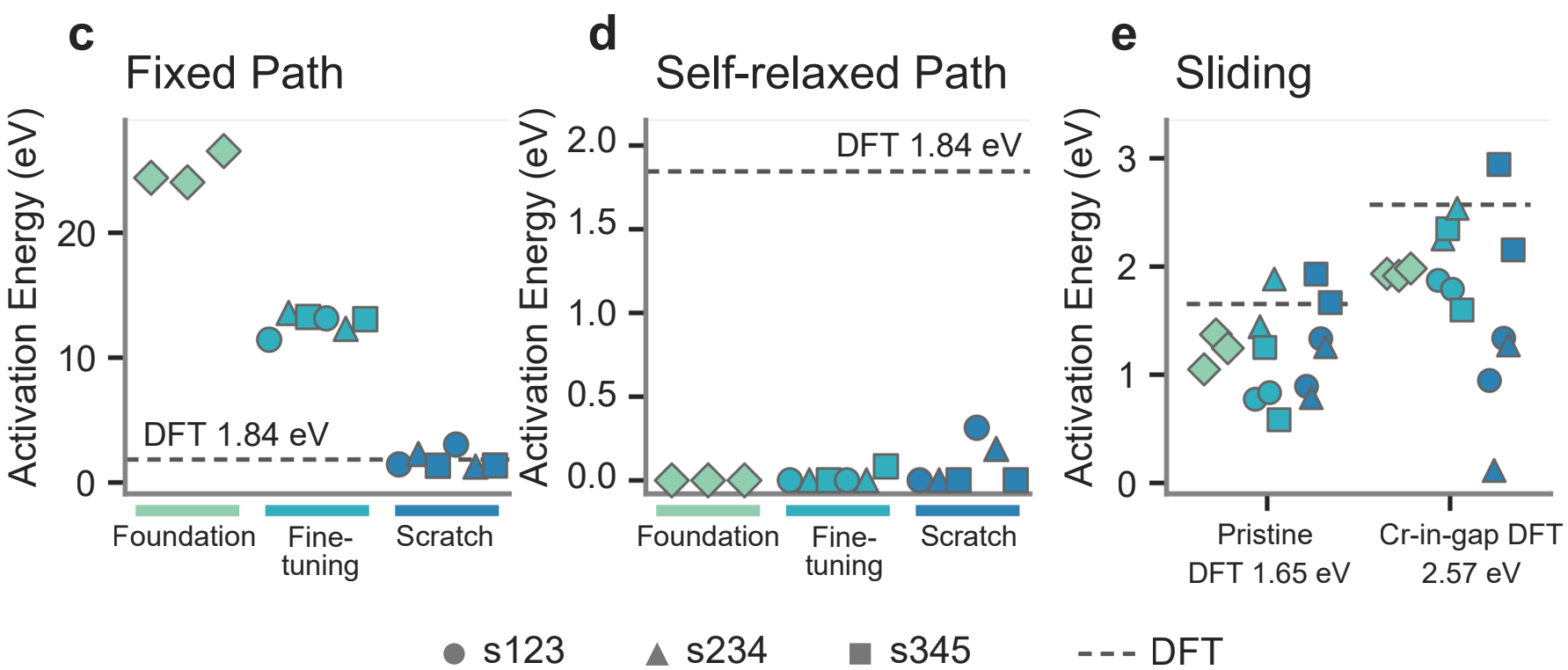


Foundation Fine-tuning Scratch ○ 600 K ● Multi-T ○ seed 123 △ seed 234 □ seed 345


a
in-gap diffusion
Fixed path
Activation Energy (eV)
Reaction Coordinate
b
gap→QL
Relative Energy (eV)
Reaction Coordinate
c
within-QL
Relative Energy (eV)
Reaction Coordinate
d
in-gap diffusion
Self-relaxted
Activation Energy (eV)
Reaction Coordinate
e
gap→QL
Relative Energy (eV)
Reaction Coordinate
f
within-QL
Relative Energy (eV)
Reaction Coordinate
Foundation
Fine-tuning · 600 K
Fine-tuning · Multi-T
Scratch · 600 K
Scratch · Multi-T
DFT-NEB


a
Training strategies
Foundation
Scratch
Fine-tuning
Energy prediction
Evaluation tasks
Fixed-path migration
Self-relaxed migration
Sliding
Δx
$E_\theta(q_s^{ref})$
$E_\theta(q_s^{\theta})$
$E_\theta(\Delta x)$
in-gap
gap -> QL
within QL
pristine
Cr-in-gap


b
Fixed-path migration
Self-relaxed migration
Sliding
in-gap
gap -> QL
within QL
in-gap
gap -> QL
within QL
pristine
Cr-in-gap
Foundation
MP-0B3
OMAT-0
MPA0
Fine-tuning
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Scratch
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Training strategy
Foundation
Fine-tuning
Scratch
Absolute error (eV)
0
0.1
1
5
25
Predicted energy (eV)
27
10
5
2
0.5
0.1
0


c
Fixed Path
Activation Energy (eV)
20
10
0
DFT 1.84 eV
Foundation
Fine-tuning
Scratch
d
Self-relaxed Path
Activation Energy (eV)
2.0
1.5
1.0
0.5
0.0
DFT 1.84 eV
Foundation
Fine-tuning
Scratch
e
Sliding
Activation Energy (eV)
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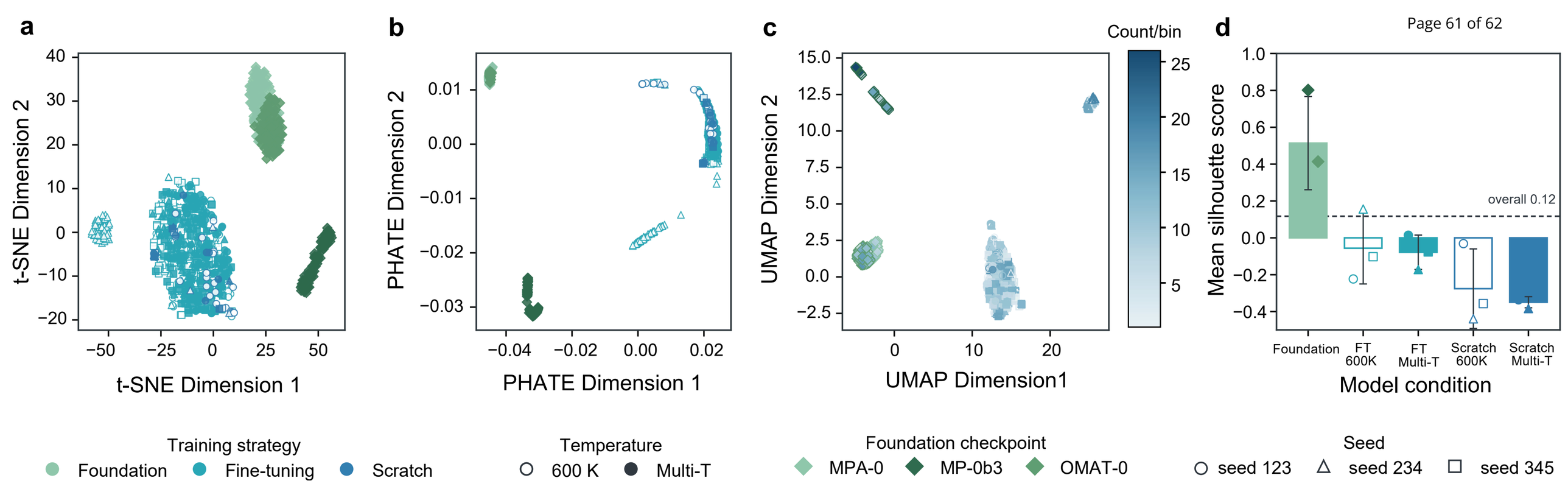
2
1
0
Pristine
DFT 1.65 eV
Cr-in-gap DFT
2.57 eV
s123
s234
s345
DFT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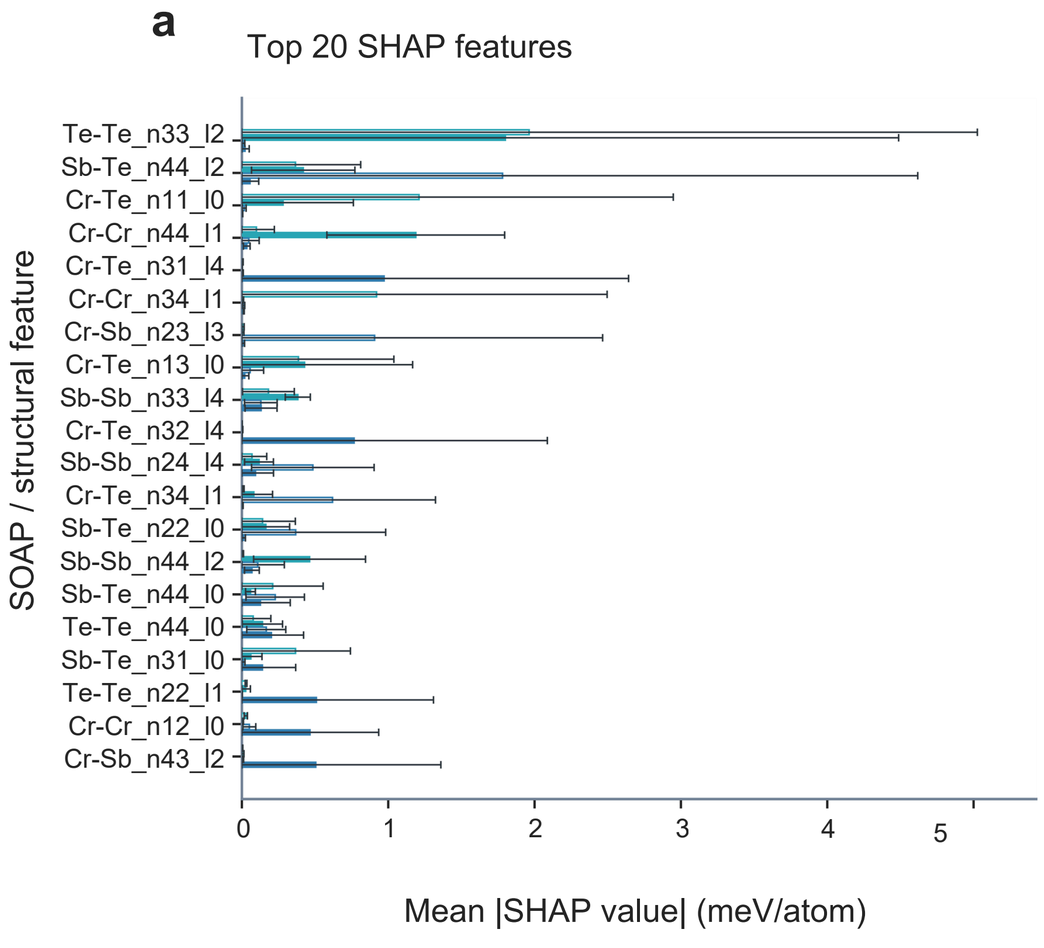
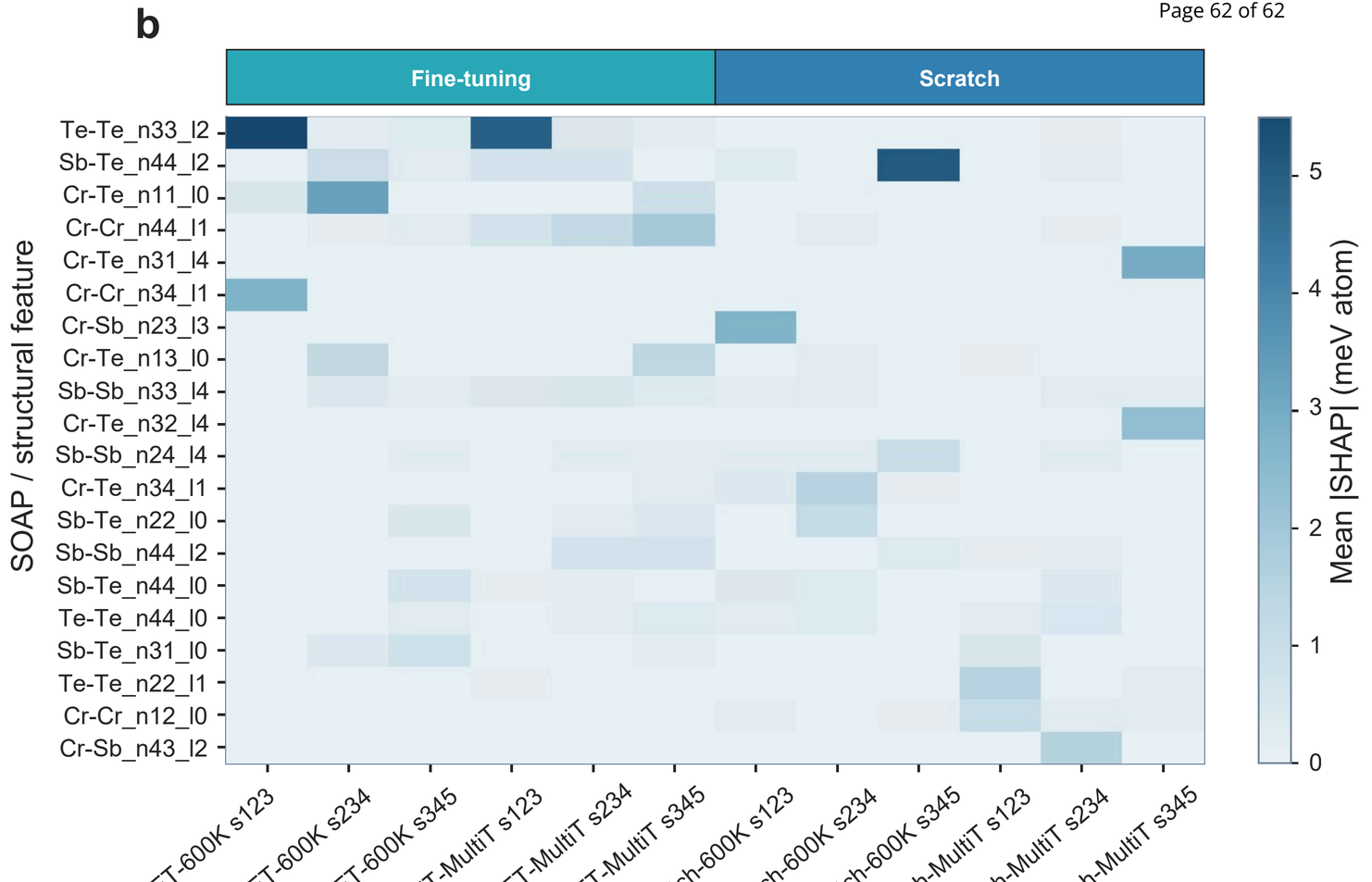
t-SNE Dimension 2
t-SNE Dimension 1
b
PHATE Dimension 2
PHATE Dimension 1
c
Count/bin
UMAP Dimension 2
UMAP Dimension1
d
Mean silhouette score
overall 0.12
Foundation
FT 600K
FT Multi-T
Scratch 600K
Scratch Multi-T
Model condition
Training strategy
Foundation
Fine-tuning
Scratch
Temperature
600 K
Multi-T
Foundation checkpoint
MPA-0
MP-0b3
OMAT-0
Seed
seed 123
seed 234
seed 345


a
Top 20 SHAP features
Te-Te_n33_l2
Sb-Te_n44_l2
Cr-Te_n11_l0
Cr-Cr_n44_l1
Cr-Te_n31_l4
Cr-Cr_n34_l1
Cr-Sb_n23_l3
Cr-Te_n13_l0
Sb-Sb_n33_l4
Cr-Te_n32_l4
Sb-Sb_n24_l4
Cr-Te_n34_l1
Sb-Te_n22_l0
Sb-Sb_n44_l2
Sb-Te_n44_l0
Te-Te_n44_l0
Sb-Te_n31_l0
Te-Te_n22_l1
Cr-Cr_n12_l0
Cr-Sb_n43_l2
SOAP / structural feature
0
1
2
3
4
5
Mean |SHAP value| (meV/atom)
Fine-tuning
Scratch


b
Fine-tuning
Scratch
Te-Te_n33_l2
Sb-Te_n44_l2
Cr-Te_n11_l0
Cr-Cr_n44_l1
Cr-Te_n31_l4
Cr-Cr_n34_l1
Cr-Sb_n23_l3
Cr-Te_n13_l0
Sb-Sb_n33_l4
Cr-Te_n32_l4
Sb-Sb_n24_l4
Cr-Te_n34_l1
Sb-Te_n22_l0
Sb-Sb_n44_l2
Sb-Te_n44_l0
Te-Te_n44_l0
Sb-Te_n31_l0
Te-Te_n22_l1
Cr-Cr_n12_l0
Cr-Sb_n43_l2
SOAP / structural feature
FT-600K s123
FT-600K s234
FT-600K s345
FT-MultiT s123
FT-MultiT s234
FT-MultiT s345
Scratch-600K s123
Scratch-600K s234
Scratch-600K s345
Scratch-MultiT s123
Scratch-MultiT s234
Scratch-MultiT s345
0
1
2
3
4
5
Mean |SHAP| (meV atom)
600 K
Multi-T